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Medical device and method for detecting electrical signal noise

Abstract

A medical device is configured to sense event signals from a cardiac electrical signal and determine maximum amplitudes of cardiac electrical signal segments associated with sensed event signals. The medical device is configured to determine at least one tachyarrhythmia metric based on at least a greatest one of the determined maximum amplitudes. The medical device may determine when the at least one tachyarrhythmia metric does not meet true tachyarrhythmia evidence and, in response, determine when the maximum amplitudes meet suspected noise criteria. The medical device may withhold a tachyarrhythmia detection and tachyarrhythmia therapy when suspected noise criteria are met.

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Primary Examiner: Johnson; Nicole F

Background/Summary

REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of U.S. patent application Ser. No. 16/874,920, filed on May 15, 2020, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

(1) The disclosure relates generally to a medical device and method for detecting electrical signal noise.

BACKGROUND

(2) Medical devices may sense electrophysiological signals from the heart, brain, nerve, muscle or other tissue. Such devices may be implantable, wearable or external devices using implantable

and/or surface (skin) electrodes for sensing the electrophysiological signals. In various medical devices or medical device systems, implantable, transcutaneous, or cutaneous (skin) electrodes may be positioned for sensing an electrophysiological signal by the medical device, which may be an implantable, external or wearable medical device. Such devices may include devices configured to monitor an electrophysiological signal for a medical condition or health purposes (including, but not limited to fitness trackers, watches, or other medical or fitness devices).

(3) In some cases, such devices may be configured to deliver a therapy based on the sensed electrophysiological signals. For example, implantable or external cardiac pacemakers, cardioverter defibrillators, cardiac monitors and the like, sense cardiac electrical signals from a patient's heart. A cardiac pacemaker or cardioverter defibrillator may deliver therapeutic electrical stimulation to the heart via electrodes carried by one or more medical electrical leads and/or electrodes on a housing of the medical device. The electrical stimulation may include signals such as pacing pulses or cardioversion or defibrillation shocks. In some cases, a medical device may sense cardiac electrical signals attendant to the intrinsic or pacing-evoked depolarizations of the myocardium and control delivery of stimulation signals to the heart based on the sensed cardiac electrical signals attendant to the myocardial depolarizations. Upon detection of an abnormal rhythm, such as bradycardia, tachycardia or fibrillation, an appropriate electrical stimulation signal or signals may be delivered to restore or maintain a more normal rhythm of the heart. For example, an implantable cardioverter defibrillator (ICD) may deliver pacing pulses to the heart of the patient upon detecting bradycardia or tachycardia or deliver cardioversion or defibrillation (CV/DF) shocks to the heart upon detecting tachycardia or fibrillation.

(4) A medical device may sense cardiac electrical signals from a heart chamber and deliver electrical stimulation therapies to the heart chamber using electrodes carried by a transvenous medical electrical lead. Cardiac signals sensed within a heart chamber using endocardial electrodes, for example, generally have a high signal strength and quality for reliably sensing near-field cardiac electrical events, such as ventricular R-waves sensed from within a ventricle, but may still be corrupted by noise such as electromagnetic interference (EMI), noise signals due to lead issues, or other non-cardiac electrical signal noise. In some proposed or available ICD systems, an extra-cardiac lead may be coupled to the ICD, in which case cardiac signal sensing from outside the heart may present challenges in reliably sensing cardiac electrical event signals, particularly in the presence of environmental or non-cardiac electrical signal noise.

SUMMARY

(5) In general, the disclosure is directed to a medical device and techniques for determining noise corruption of an electrophysiological signal sensed by the medical device. The techniques disclosed herein may be used in conjunction with a variety of cardiac monitoring and/or therapy delivery devices, including devices that monitor a patient heart rate for detecting tachyarrhythmia. For example, a determination of noise present in a cardiac electrical signal may be included in heart rate monitoring and tachyarrhythmia detection methods to avoid false tachyarrhythmia detection due to the presence electrical noise, such as electromagnetic interference (EMI), non-cardiac myopotential signals or other electrical noise signals. A device operating according to the techniques disclosed herein may determine when the amplitude of signals associated with event signals sensed as cardiac events meet suspected noise criteria. The medical device may determine a tachyarrhythmia metric based on at least a greatest maximum amplitude signal. The medical device may determine one or more interval-based tachyarrhythmia metrics and/or one or more morphology-based tachyarrhythmia metrics. The medical device may determine when true tachyarrhythmia evidence criteria are met based on the tachyarrhythmia metric(s). When the true tachyarrhythmia evidence criteria are unmet and the suspected noise criteria are met, the medical device may determine noise corruption of the cardiac electrical signal. Determination of noise may be used to withhold a tachyarrhythmia detection by the medical device even when other tachyarrhythmia detection criteria, such as rate based criteria, are met. In this way, delivery of

unnecessary therapy, such as a CV/DF shock, may be avoided when tachyarrhythmia might otherwise be falsely detected due to noise corruption of the cardiac electrical signal.

(6) In some examples, a medical device as disclosed herein may be configured to detect ventricular tachyarrhythmia, e.g., ventricular tachycardia (VT) or ventricular fibrillation (VF), based on detecting a ventricular rate that is faster than a tachyarrhythmia detection rate for at least a predetermined number of ventricular cycles. The VT or VF rate may be detected by sensing R-waves from a cardiac electrical signal, determining ventricular intervals or RR intervals (RRIs) between consecutively sensed R-waves, and counting the number of ventricular intervals that are shorter than VT or VF detection intervals. Non-cardiac noise may be oversensed as ventricular R-waves due to cardiac signal amplitude variability and/or due to episodes of non-cardiac noise. Oversensing of non-cardiac noise may cause the medical device to falsely increase the count of VT or VF intervals when an underlying normal sinus rhythm may be present. A medical device operating according to the techniques disclosed herein may determine noise corruption of a cardiac electrical signal, which may be occurring during a series of ventricular intervals that include tachyarrhythmia detection intervals.

(7) In one example, the disclosure provides a medical device including a sensing circuit and a control circuit coupled to the sensing circuit. The sensing circuit may be configured to sense at least one cardiac electrical signal and sense event signals from the at least one cardiac electrical signal. The control circuit may be configured to determine a maximum amplitude associated with each one of multiple sensed event signals, identify a greatest maximum amplitude from the determined maximum amplitudes and determine at least one tachyarrhythmia metric based on at least the identified greatest maximum amplitude. The control circuit may be further configured to determine the at least one tachyarrhythmia metric does not meet true tachyarrhythmia evidence criteria and, in response to the at least one tachyarrhythmia metric not meeting true tachyarrhythmia evidence criteria, determine that suspected noise criteria are met based on the determined maximum amplitudes. The control circuit may determine that a tachyarrhythmia detection criterion is met based on event signals sensed by the sensing circuit and, in response to determining that suspected noise criteria are met, withhold a tachyarrhythmia detection when the tachyarrhythmia detection criterion is determined to be met.

(8) In another example, the disclosure provides a method including sensing at least one cardiac electrical signal, sensing event signals from the at least one cardiac electrical signal and determining a maximum amplitude associated with each one of multiple sensed event signals. The method may further include identifying a greatest maximum amplitude from the determined maximum amplitudes, determining at least one tachyarrhythmia metric based on at least the identified greatest maximum amplitude, and determining the at least one tachyarrhythmia metric does not meet true tachyarrhythmia evidence criteria. In response to the at least one tachyarrhythmia metric not meeting true tachyarrhythmia evidence criteria, the method may include determining that suspected noise criteria are met based on the determined maximum amplitudes. The method may further include determining that a tachyarrhythmia detection criterion is met based on the plurality of sensed event signals. In response to determining that suspected noise criteria are met, the method may include withholding a tachyarrhythmia detection when the tachyarrhythmia detection criterion is determined to be met.

(9) In another example, the disclosure provides a non-transitory computer-readable medium storing a set of instructions which, when executed by a control circuit of a medical device, cause the medical device to sense at least one cardiac electrical signal, sense event signals from the at least one cardiac electrical signal, determine a maximum amplitude associated with each of a plurality of the sensed event signals, identify a greatest maximum amplitude from the determined maximum amplitudes and determine at least one tachyarrhythmia metric based on at least the identified greatest maximum amplitude. The instructions further cause the device to determine the at least one tachyarrhythmia metric does not meet true tachyarrhythmia evidence criteria and, in response to the

at least one tachyarrhythmia metric not meeting true tachyarrhythmia evidence criteria, determine that suspected noise criteria are met based on the determined maximum amplitudes. The instructions may cause the device to determine that a tachyarrhythmia detection criterion is met based on the sensed event signals and, in response to determining that suspected noise criteria are met, withhold a tachyarrhythmia detection when the tachyarrhythmia detection criterion is determined to be met.

(10) This summary is intended to provide an overview of the subject matter described in this disclosure. It is not intended to provide an exclusive or exhaustive explanation of the apparatus and methods described in detail within the accompanying drawings and description below. Further details of one or more examples are set forth in the accompanying drawings and the description below.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIGS. 1A and 1B are conceptual diagrams of an extra-cardiovascular ICD system configured to sense cardiac electrical events and deliver cardiac electrical stimulation therapies according to one example.
- (2) FIGS. 2A-2C are conceptual diagrams of a patient implanted with an extra-cardiovascular ICD system in a different implant configuration than the arrangement shown in FIGS. 1A-1B.
- (3) FIG. 3 is a conceptual diagram of an ICD according to one example.
- (4) FIG. 4 is a diagram of circuitry that may be included in a sensing circuit of the ICD of FIG. 3.
- (5) FIG. 5 is a flow chart of a method for determining that a cardiac electrical signal is corrupted by noise according to one example.
- (6) FIG. 6 is diagram of a first cardiac electrical signal and a second cardiac electrical signal that may be used for sensing cardiac events and determining noise corruption according to one example.
- (7) FIG. 7 is a flow chart of a method for determining when tachyarrhythmia evidence criteria are not met for enabling a determination of suspected noise corruption according to one example.
- (8) FIG. 8 is a diagram of a cardiac electrical signal segment stored in response to an R-wave sensed event signal.
- (9) FIG. 9 is a flow chart of a method for determining a gross morphology amplitude metric of a cardiac electrical signal segment for detecting a tachyarrhythmia morphology according to one example.
- (10) FIG. 10 is a flow chart of a method for detecting a tachyarrhythmia morphology in a cardiac electrical signal segment based on a gross morphology signal width metric according to one example.
- (11) FIG. 11 is a flow chart of a method for controlling tachyarrhythmia detection and therapy by a medical device according to one example.
- (12) FIG. 12 is a flow chart of a method performed by a medical device for detecting non-cardiac noise corruption of a cardiac signal and rejecting a ventricular tachyarrhythmia detection in response to determining noise corruption according to another example.

DETAILED DESCRIPTION

(13) In general, this disclosure describes a medical device and techniques for determining the presence of noise in an electrical signal sensed by the medical device, such as a cardiac electrical signal. In some examples, the medical device may be configured to sense cardiac electrical events attendant to myocardial depolarizations, e.g., atrial P-waves attendant to atrial myocardial depolarizations and/or ventricular R-waves attendant to ventricular myocardial depolarizations, from the cardiac electrical signal. The medical device may determine the heart rate or rhythm and a

need for therapy delivery based on the sensed cardiac electrical event signals. For example, atrial or ventricular tachyarrhythmia may be detected by the medical device based on sensed cardiac electrical event signals. In some examples, the medical device may be configured to sense R-waves attendant to ventricular depolarizations from a cardiac electrical signal for use in controlling ventricular pacing and detecting ventricular tachyarrhythmias. A ventricular tachyarrhythmia may be detected in response to sensing a threshold number of R-waves occurring at a time interval from a preceding R-wave that is less than a tachyarrhythmia detection interval.

(14) Non-cardiac electrical noise present in the cardiac signal, e.g., electromagnetic interference (EMI) or skeletal muscle myopotential signals, may be oversensed as R-waves, resulting in false, short RRIs being determined as ventricular tachyarrhythmia intervals. In some instances, variability in the R-wave signal strength due to patient motion or other factors may result in oversensing of non-cardiac noise, leading to relatively short RRIs being counted toward tachyarrhythmia detection when the underlying rhythm may actually be a normal sinus rhythm. False tachyarrhythmia detection may lead to a CV/DF shock or other tachyarrhythmia therapy delivered by the medical device, such as anti-tachyarrhythmia pacing (ATP), when a therapy may not be needed.

(15) In some examples, the medical device performing the techniques disclosed herein may be included in an extra-cardiovascular ICD system. As used herein, the term “extra-cardiovascular” refers to a position outside the heart and/or blood vessels and/or pericardium surrounding the heart of a patient. Implantable electrodes carried by extra-cardiovascular leads may be positioned extra-thoracically (outside the ribcage and sternum) or intra-thoracically (beneath the ribcage or sternum) but generally not in intimate contact with myocardial tissue. The medical device may be coupled to a lead in an “extra-cardiac” location that positions electrodes within a blood vessel(s) not surrounding the heart. For example, a medical lead may be advanced along a venous pathway to position electrodes within the internal thoracic vein (ITV), an intercostal vein, the superior epigastric vein, or the azygos, hemizygous, or accessory hemizygous veins, as examples.

(16) Patient positional changes or patient physical activity as well as other factors may lead to variation in the cardiac event signal amplitudes, e.g., P-wave amplitudes, R-wave amplitudes and T-wave amplitudes, in the signal sensed from an extra-cardiovascular or extra-cardiac location. Furthermore, the presence and amplitude of non-cardiac noise in a cardiac electrical signal may be highly variable. Cardiac signals sensed via extra-cardiovascular or extra-cardiac electrodes may be more susceptible to signal amplitude variability and noise contamination, e.g., due to myopotentials or environmental EMI, than cardiac signals sensed using transvenous intracardiac electrodes or epicardial electrodes. However, the techniques disclosed herein for determining noise corruption of a cardiac electrical signal may be implemented in conjunction with a variety of electrode configurations used for sensing cardiac electrical signals.

(17) The medical device and techniques disclosed herein provide a method for determining the presence of noise in a cardiac electrical signal and withholding detection of a tachyarrhythmia when noise is determined to be present. The illustrative examples presented herein involve sensing cardiac electrical signals for the detection of ventricular tachyarrhythmia. The disclosed techniques, however, may be implemented in a medical device configured to sense atrial and/or ventricular cardiac events for detecting a variety of cardiac rhythms, such as bradycardia, tachycardia, fibrillation, etc. For example, a cardiac device using the disclosed techniques may be configured to sense P-waves, e.g., for detecting (and optionally treating) atrial tachyarrhythmia. In this case, the medical device may count PP intervals occurring between consecutively sensed atrial P-waves which are less than an atrial tachyarrhythmia detection interval. Cardiac electrical signals, which may be sensed from within or outside an atrial chamber, may be analyzed for determining noise corruption of the cardiac electrical signal using the techniques disclosed herein. An atrial tachyarrhythmia episode may be rejected or withheld based on a determination of noise corruption.

(18) More generally, the disclosed techniques may be adapted for use in any device that is configured to determine a heart rate from sensed cardiac electrical signals, such as fitness trackers,

watches, or other heart rate monitors. When the cardiac electrical signal is corrupted by non-cardiac noise, the determined heart rate may be incorrect, e.g., over-estimated, due to the non-cardiac noise signals being falsely sensed as cardiac events.

(19) FIGS. 1A and 1B are conceptual diagrams of an ICD system 10 configured to sense cardiac electrical events and deliver cardiac electrical stimulation therapies according to one example. FIG. 1A is a front view of ICD system 10 implanted within patient 12. FIG. 1B is a side view of ICD system 10 implanted within patient 12. ICD system 10 includes an ICD 14 connected to an extra-cardiovascular electrical stimulation and sensing lead 16. FIGS. 1A and 1B are described in the context of an ICD system 10 capable of providing high voltage CV/DF shocks, and in some examples cardiac pacing pulses, in response to detecting a cardiac tachyarrhythmia.

(20) ICD 14 includes a housing 15 that forms a hermetic seal that protects internal components of ICD 14. The housing 15 of ICD 14 may be formed of a conductive material, such as titanium or titanium alloy. The housing 15 may function as an electrode (sometimes referred to as a “can” electrode). Housing 15 may be used as an active can electrode for use in delivering CV/DF shocks or other high voltage pulses delivered using a high voltage therapy circuit. In other examples, housing 15 may be available for use in delivering unipolar, low voltage cardiac pacing pulses and/or for sensing cardiac electrical signals in combination with electrodes carried by lead 16. In other instances, the housing 15 of ICD 14 may include a plurality of electrodes on an outer portion of the housing. The outer portion(s) of the housing 15 functioning as an electrode(s) may be coated with a material, such as titanium nitride, e.g., for reducing post-stimulation polarization artifact.

(21) ICD 14 includes a connector assembly 17 (also referred to as a connector block or header) that includes electrical feedthroughs crossing housing 15 to provide electrical connections between conductors extending within the lead body 18 of lead 16 and electronic components included within the housing 15 of ICD 14. As will be described in further detail herein, housing 15 may house one or more processors, memories, transceivers, cardiac electrical signal sensing circuitry, therapy delivery circuitry, power sources and other components for sensing cardiac electrical signals, detecting a heart rhythm, and controlling and delivering electrical stimulation pulses to treat an abnormal heart rhythm.

(22) Elongated lead body 18 has a proximal end 27 that includes a lead connector (not shown) configured to be connected to ICD connector assembly 17 and a distal portion 25 that includes one or more electrodes. In the example illustrated in FIGS. 1A and 1B, the distal portion 25 of lead body 18 includes defibrillation electrodes 24 and 26 and pace/sense electrodes 28 and 30. In some cases, defibrillation electrodes 24 and 26 may together form a defibrillation electrode in that they may be configured to be activated concurrently. Alternatively, defibrillation electrodes 24 and 26 may form separate defibrillation electrodes in which case each of the electrodes 24 and 26 may be activated independently.

(23) Electrodes 24 and 26 (and in some examples housing 15) are referred to herein as defibrillation electrodes because they are utilized, individually or collectively, for delivering high voltage stimulation therapy (e.g., cardioversion or defibrillation shocks). Electrodes 24 and 26 may be elongated coil electrodes and generally have a relatively high surface area for delivering high voltage electrical stimulation pulses compared to pacing and sensing electrodes 28 and 30. However, electrodes 24 and 26 and housing 15 may also be utilized to provide pacing functionality, sensing functionality or both pacing and sensing functionality in addition to or instead of high voltage stimulation therapy. In this sense, the use of the term “defibrillation electrode” herein should not be considered as limiting the electrodes 24 and 26 for use in only high voltage cardioversion/defibrillation shock therapy applications. For example, either of electrodes 24 and 26 may be used as a sensing electrode in a sensing vector for sensing cardiac electrical signals and determining a need for an electrical stimulation therapy.

(24) Electrodes 28 and 30 are relatively smaller surface area electrodes which are available for use in sensing electrode vectors for sensing cardiac electrical signals and may be used for delivering

relatively low voltage pacing pulses in some configurations. Electrodes **28** and **30** are referred to as pace/sense electrodes because they are generally configured for use in low voltage applications, e.g., used as either a cathode or anode for delivery of pacing pulses and/or sensing of cardiac electrical signals, as opposed to delivering high voltage CV/DF shocks. In some instances, electrodes **28** and **30** may provide only pacing functionality, only sensing functionality or both.

(25) ICD **14** may sense cardiac electrical signals corresponding to electrical activity of heart **8** via a combination of sensing electrode vectors that include combinations of electrodes **24**, **26**, **28** and/or **30**. In some examples, housing **15** of ICD **14** is used in combination with one or more of electrodes **24**, **26**, **28** and/or **30** in a sensing electrode vector. Various sensing electrode vectors utilizing combinations of electrodes **24**, **26**, **28**, and **30** and housing **15** may be selected sensing at least one cardiac electrical signal and sensing cardiac events from the cardiac electrical signal. In some example, two or more sensing electrode vectors may be selected from the available electrodes for sensing two or more cardiac electrical signals using the respective sensing electrode vectors.

(26) In the example illustrated in FIGS. **1A** and **1B**, electrode **28** is located proximal to defibrillation electrode **24**, and electrode **30** is located between defibrillation electrodes **24** and **26**. One, two or more pace/sense electrodes may be carried by lead body **18**. For instance, a third pace/sense electrode may be located distal to defibrillation electrode **26** in some examples. Electrodes **28** and **30** are illustrated as ring electrodes; however, electrodes **28** and **30** may comprise any of a number of different types of electrodes, including ring electrodes, short coil electrodes, hemispherical electrodes, directional electrodes, segmented electrodes, or the like. Electrodes **28** and **30** may be positioned at other locations along lead body **18** and are not limited to the positions shown. In other examples, lead **16** may include fewer or more pace/sense electrodes and/or defibrillation electrodes than the example shown here.

(27) In the example shown, lead **16** extends subcutaneously or submuscularly over the ribcage **32** medially from the connector assembly **27** of ICD **14** toward a center of the torso of patient **12**, e.g., toward xiphoid process **20** of patient **12**. At a location near xiphoid process **20**, lead **16** bends or turns and extends superiorly, subcutaneously or submuscularly, over the ribcage and/or sternum, substantially parallel to sternum **22**. Although illustrated in FIG. **1A** as being offset laterally from and extending substantially parallel to sternum **22**, the distal portion **25** of lead **16** may be implanted at other locations, such as over sternum **22**, offset to the right or left of sternum **22**, angled laterally from sternum **22** toward the left or the right, or the like. Alternatively, lead **16** may be placed along other subcutaneous or submuscular paths. The path of extra-cardiovascular lead **16** may depend on the location of ICD **14**, the arrangement and position of electrodes carried by the lead body **18**, and/or other factors. The techniques disclosed herein are not limited to a particular path of lead **16** or final locations of electrodes **24**, **26**, **28** and **30** carried by a lead for sensing cardiac electrical signals.

(28) Electrical conductors (not illustrated) extend through one or more lumens of the elongated lead body **18** of lead **16** from the lead connector at the proximal lead end **27** to electrodes **24**, **26**, **28**, and **30** located along the distal portion **25** of the lead body **18**. The elongated electrical conductors contained within the lead body **18**, which may be separate respective insulated conductors within the lead body **18**, are each electrically coupled with respective defibrillation electrodes **24** and **26** and pace/sense electrodes **28** and **30**. The respective conductors electrically couple the electrodes **24**, **26**, **28**, and **30** to circuitry, such as a therapy delivery circuit and/or a sensing circuit, of ICD **14** via connections in the connector assembly **17**, including associated electrical feedthroughs crossing housing **15**. The electrical conductors transmit therapy from a therapy delivery circuit within ICD **14** to one or more of defibrillation electrodes **24** and **26** and/or pace/sense electrodes **28** and **30** and transmit sensed electrical signals produced by the patient's heart **8** from one or more of defibrillation electrodes **24** and **26** and/or pace/sense electrodes **28** and **30** to the sensing circuit within ICD **14**.

(29) The lead body **18** of lead **16** may be formed from a non-conductive material, including

silicone, polyurethane, fluoropolymers, mixtures thereof, and/or other appropriate materials, and shaped to form one or more lumens within which the one or more conductors extend. Lead body **18** may be tubular or cylindrical in shape. In other examples, the distal portion **25** (or all of) the elongated lead body **18** may have a flat, ribbon or paddle shape. Lead body **18** may be formed having a preformed distal portion **25** that is generally straight, curving, bending, serpentine, undulating or zig-zagging.

(30) In the example shown, lead body **18** includes a curving distal portion **25** having two “C” shaped curves, which together may resemble the Greek letter epsilon, “ε.” Defibrillation electrodes **24** and **26** are each carried by one of the two respective C-shaped portions of the lead body distal portion **25**. The two C-shaped curves are seen to extend or curve in the same direction away from a central axis of lead body **18**, along which pace/sense electrodes **28** and **30** are positioned. Pace/sense electrodes **28** and **30** may, in some instances, be approximately aligned with the central axis of the straight, proximal portion of lead body **18** such that mid-points of defibrillation electrodes **24** and **26** are laterally offset from pace/sense electrodes **28** and **30**.

(31) Other examples of extra-cardiovascular leads including one or more defibrillation electrodes and one or more pacing and sensing electrodes carried by curving, serpentine, undulating or zig-zagging distal portion of the lead body **18** that may be implemented with the techniques described herein are generally disclosed in pending U.S. Pat. publication No. 2016/0158567 (Marshall, et al.), incorporated herein by reference in its entirety. The techniques disclosed herein are not limited to any particular lead body design, however. In other examples, lead body **18** is a flexible elongated lead body without any pre-formed shape, bends or curves.

(32) ICD **14** analyzes the cardiac electrical signals received from one or more sensing electrode vectors to monitor for abnormal rhythms, such as bradycardia, ventricular tachycardia (VT) or ventricular fibrillation (VF). ICD **14** may analyze the heart rate and morphology of the cardiac electrical signals to monitor for tachyarrhythmia in accordance with any of a number of tachyarrhythmia detection techniques. Example techniques for detecting a tachyarrhythmia are described in conjunction with the flow charts and diagrams presented herein.

(33) ICD **14** generates and delivers electrical stimulation therapy in response to detecting a tachyarrhythmia (e.g., VT or VF) using a therapy delivery electrode vector which may be selected from any of the available electrodes **24**, **26**, **28** **30** and/or housing **15**. ICD **14** may deliver ATP in response to VT detection and in some cases may deliver ATP prior to a CV/DF shock or during high voltage capacitor charging in an attempt to avert the need for delivering a CV/DF shock. If ATP does not successfully terminate VT or when VF is detected, ICD **14** may deliver one or more CV/DF shocks via one or both of defibrillation electrodes **24** and **26** and/or housing **15**. ICD **14** may deliver the CV/DF shocks using electrodes **24** and **26** individually or together as a cathode (or anode) and with the housing **15** as an anode (or cathode). ICD **14** may generate and deliver other types of electrical stimulation pulses such as post-shock pacing pulses, asystole pacing pulses, or bradycardia pacing pulses using a pacing electrode vector that includes one or more of the electrodes **24**, **26**, **28**, and **30** and the housing **15** of ICD **14**.

(34) ICD **14** is shown implanted subcutaneously on the left side of patient **12** along the ribcage **32**. ICD **14** may, in some instances, be implanted between the left posterior axillary line and the left anterior axillary line of patient **12**. ICD **14** may, however, be implanted at other subcutaneous or submuscular locations in patient **12**. For example, ICD **14** may be implanted in a subcutaneous pocket in the pectoral region. In this case, lead **16** may extend subcutaneously or submuscularly from ICD **14** toward the manubrium of sternum **22** and bend or turn and extend inferiorly from the manubrium to the desired location subcutaneously or submuscularly. In yet another example, ICD **14** may be placed abdominally. Lead **16** may be implanted in other extra-cardiovascular or extra-cardiac locations as well. For instance, as described with respect to FIGS. 2A-2C, the distal portion **25** of lead **16** may be implanted underneath the sternum/ribcage in the substernal space. FIGS. 1A and 1B are illustrative in nature and should not be considered limiting of the practice of the

techniques disclosed herein.

(35) An external device **40** is shown in telemetric communication with ICD **14** by a communication link **42**. External device **40** may include a processor **52**, memory **53**, display **54**, user interface **56** and telemetry unit **58**. Processor **52** controls external device operations and processes data and signals received from ICD **14**. Display **54**, which may include a graphical user interface, displays data and other information to a user for reviewing ICD operation and programmed parameters as well as cardiac electrical signals retrieved from ICD **14**.

(36) User interface **56** may include a mouse, touch screen, key pad or the like to enable a user to interact with external device **40** to initiate a telemetry session with ICD **14** for retrieving data from and/or transmitting data to ICD **14**, including programmable parameters for controlling cardiac event sensing and therapy delivery. Telemetry unit **58** includes a transceiver and antenna configured for bidirectional communication with a telemetry circuit included in ICD **14** and is configured to operate in conjunction with processor **52** for sending and receiving data relating to ICD functions via communication link **42**.

(37) Communication link **42** may be established between ICD **14** and external device **40** using a radio frequency (RF) link such as BLUETOOTH®, Wi-Fi, or Medical Implant Communication Service (MICS) or other RF or communication frequency bandwidth or communication protocols. Data stored or acquired by ICD **14**, including physiological signals or associated data derived therefrom, results of device diagnostics, and histories of detected rhythm episodes and delivered therapies, may be retrieved from ICD **14** by external device **40** following an interrogation command.

(38) External device **40** may be embodied as a programmer used in a hospital, clinic or physician's office to retrieve data from ICD **14** and to program operating parameters and algorithms in ICD **14** for controlling ICD functions. External device **40** may alternatively be embodied as a home monitor or hand held device. External device **40** may be used to program cardiac signal sensing parameters, cardiac rhythm detection parameters and therapy control parameters used by ICD **14**. At least some control parameters used in detecting noise according to techniques disclosed herein may be programmed into ICD **14** using external device **40** in some examples.

(39) FIGS. 2A-2C are conceptual diagrams of patient **12** implanted with extra-cardiovascular ICD system **10** in a different implant configuration than the arrangement shown in FIGS. 1A-1B. FIG. 2A is a front view of patient **12** implanted with ICD system **10**. FIG. 2B is a side view of patient **12** implanted with ICD system **10**. FIG. 2C is a transverse view of patient **12** implanted with ICD system **10**. In this arrangement, extra-cardiovascular lead **16** of system **10** is implanted at least partially underneath sternum **22** of patient **12**. Lead **16** extends subcutaneously or submuscularly from ICD **14** toward xiphoid process **20** and at a location near xiphoid process **20** bends or turns and extends superiorly within anterior mediastinum **36** in a substernal position.

(40) Anterior mediastinum **36** may be viewed as being bounded laterally by pleurae **39**, posteriorly by pericardium **38**, and anteriorly by sternum **22** (see FIG. 2C) and may include loose connective tissue (such as areolar tissue), some lymph vessels, lymph glands, substernal musculature (e.g., transverse thoracic muscle), branches of the internal thoracic artery, and the internal thoracic vein. The distal portion **25** of lead **16** may extend along the posterior side of sternum **22** substantially within the anterior mediastinum **36**. A lead implanted such that the distal portion **25** is substantially within anterior mediastinum **36**, may be referred to as a “substernal lead.”

(41) In the example illustrated in FIGS. 2A-2C, lead **16** is located substantially centered under sternum **22**. In other instances, however, lead **16** may be implanted such that it is offset laterally from the center of sternum **22**. In some instances, lead **16** may extend laterally such that distal portion **25** of lead **16** is underneath/below the ribcage **32** in addition to or instead of sternum **22**. In other examples, the distal portion **25** of lead **16** may be implanted in other extra-cardiovascular, intra-thoracic locations, including the pleural cavity or around the perimeter of and adjacent to the pericardium **38** of heart **8**. For example, the distal portion **25** of lead **16** may be advanced to a

supra-diaphragmatic position, which may be within the thoracic cavity or outside the thorax in various examples. As described above, lead **16** may alternatively be advanced within a vein to position electrodes for delivering electrical stimulation pulses to heart **8** from an intravenous location.

(42) In the various example implant locations of lead **16** and electrodes **24**, **26**, **28** and **30**, cardiac signals sensed by ICD **14** may be contaminated by skeletal muscle myopotentials, environmental EMI or other non-cardiac electrical noise. Some noise signals may be oversensed as cardiac event signals, e.g., R-waves, resulting in a false heart rate being determined. A false tachyarrhythmia detection may be made, or bradycardia pacing may be withheld when it is actually needed.

Withholding a tachyarrhythmia detection when noise is suspected but the underlying rhythm is a true tachyarrhythmia may result in a therapy being withheld when it is actually needed.

Accordingly, the techniques disclosed herein provide improvements in non-cardiac noise detection by applying criteria for detecting suspected noise signals and verifying that evidence of a true tachyarrhythmia morphology and/or true cardiac event intervals is not present before determining noise corruption of the cardiac electrical signal and withholding a tachyarrhythmia detection due to noise when other tachyarrhythmia detection criteria are met. When noise is suspected but evidence of a true tachyarrhythmia is determined from the cardiac electrical signal, the suspected noise does not cause withholding of a tachyarrhythmia detection by ICD **10** when other tachyarrhythmia detection criteria are met.

(43) FIG. **3** is a conceptual diagram of ICD **14** according to one example. The electronic circuitry enclosed within housing **15** (shown schematically as an electrode in FIG. **3**) includes software, firmware and hardware that cooperatively monitor cardiac electrical signals, determine when an electrical stimulation therapy is necessary, and deliver therapy as needed according to programmed therapy delivery algorithms and control parameters. ICD **14** may be coupled to an extra-cardiovascular lead, such as lead **16** carrying extra-cardiovascular electrodes **24**, **26**, **28**, and **30**, for delivering electrical stimulation pulses to the patient's heart and for sensing cardiac electrical signals.

(44) ICD **14** includes a control circuit **80**, memory **82**, therapy delivery circuit **84**, cardiac electrical signal sensing circuit **86**, and telemetry circuit **88**. A power source **98** provides power to the circuitry of ICD **14**, including each of the components **80**, **82**, **84**, **86**, and **88** as needed. Power source **98** may include one or more energy storage devices, such as one or more rechargeable or non-rechargeable batteries. The connections between power source **98** and each of the other components **80**, **82**, **84**, **86** and **88** are to be understood from the general block diagram of FIG. **3**, but are not shown for the sake of clarity. For example, power source **98** may be coupled to one or more charging circuits included in therapy delivery circuit **84** for charging holding capacitors included in therapy delivery circuit **84** that are discharged at appropriate times under the control of control circuit **80** for producing electrical pulses according to a therapy protocol. Power source **98** is also coupled to components of cardiac electrical signal sensing circuit **86**, such as sense amplifiers, analog-to-digital converters, switching circuitry, etc. as needed.

(45) The circuits shown in FIG. **3** represent functionality included in ICD **14** and may include any discrete and/or integrated electronic circuit components that implement analog and/or digital circuits capable of producing the functions attributed to ICD **14** herein. Functionality associated with one or more circuits may be performed by separate hardware, firmware or software components, or integrated within common hardware, firmware or software components. For example, cardiac event sensing and detection of noise for rejecting sensed events or withholding detection of a tachyarrhythmia based on cardiac event intervals may be performed cooperatively by sensing circuit **86** and control circuit **80** and may include operations implemented in a processor or other signal processing circuitry included in control circuit **80** executing instructions stored in memory **82** and control signals such as blanking and timing intervals and sensing threshold amplitude signals sent from control circuit **80** to sensing circuit **86**.

(46) The various circuits of ICD **14** may include an application specific integrated circuit (ASIC), an electronic circuit, a processor (shared, dedicated, or group) and memory that execute one or more software or firmware programs, a combinational logic circuit, state machine, or other suitable components or combinations of components that provide the described functionality. The particular form of software, hardware and/or firmware employed to implement the functionality disclosed herein will be determined primarily by the particular system architecture employed in the ICD and by the particular detection and therapy delivery methodologies employed by the ICD. Providing software, hardware, and/or firmware to accomplish the described functionality in the context of any modern medical device system, given the disclosure herein, is within the abilities of one of skill in the art.

(47) Memory **82** may include any volatile, non-volatile, magnetic, or electrical non-transitory computer readable storage media, such as random access memory (RAM), read-only memory (ROM), non-volatile RAM (NVRAM), electrically-erasable programmable ROM (EEPROM), flash memory, or any other memory device. Furthermore, memory **82** may include non-transitory computer readable media storing instructions that, when executed by one or more processing circuits, cause control circuit **80** and/or other ICD components to perform various functions attributed to ICD **14** or those ICD components. The non-transitory computer-readable media storing the instructions may include any of the media listed above.

(48) Control circuit **80** communicates, e.g., via a data bus, with therapy delivery circuit **84** and sensing circuit **86** for sensing cardiac electrical activity, detecting cardiac rhythms, and controlling delivery of cardiac electrical stimulation therapies in response to sensed cardiac signals. Therapy delivery circuit **84** and sensing circuit **86** are electrically coupled to electrodes **24, 26, 28, 30** carried by lead **16** and the housing **15**, which may function as a common or ground electrode or as an active can electrode for delivering CV/DF shock pulses or cardiac pacing pulses.

(49) Cardiac electrical signal sensing circuit **86** (also referred to herein as “sensing circuit” **86**) may be selectively coupled to electrodes **28, 30** and/or housing **15** in order to monitor electrical activity of the patient's heart. Sensing circuit **86** may additionally be selectively coupled to defibrillation electrodes **24** and/or **26** for use in a sensing electrode vector together or in combination with one or more of electrodes **28, 30** and/or housing **15**. Sensing circuit **86** may be enabled to selectively receive cardiac electrical signals from one or more sensing electrode vectors from the available electrodes **24, 26, 28, 30**, and housing **15**. In some examples, at least two cardiac electrical signals from two different sensing electrode vectors may be received simultaneously by sensing circuit **86**. Sensing circuit **86** may monitor one or both of the cardiac electrical signals simultaneously for sensing cardiac electrical events and/or producing digitized cardiac signal waveforms for analysis by control circuit **80**. For example, sensing circuit **86** may include switching circuitry for selecting which of electrodes **24, 26, 28, 30**, and housing **15** are coupled to a first sensing channel **83** and which electrodes are coupled to a second sensing channel **85** of sensing circuit **86**.

(50) Each sensing channel **83** and **85** may be configured to amplify, filter and digitize the cardiac electrical signal received from selected electrodes coupled to the respective sensing channel to improve the signal quality for detecting cardiac electrical events, such as R-waves or performing other signal analysis. The cardiac event detection circuitry within sensing circuit **86** may include one or more sense amplifiers, filters, rectifiers, threshold detectors, comparators, analog-to-digital converters (ADCs), timers or other analog or digital components as described further in conjunction with FIG. **4**. A cardiac event sensing threshold may be automatically adjusted by sensing circuit **86** under the control of control circuit **80**, based on timing intervals and sensing threshold values determined by control circuit **80**, stored in memory **82**, and/or controlled by hardware, firmware and/or software of control circuit **80** and/or sensing circuit **86**.

(51) Upon detecting a cardiac event based on a sensing threshold crossing, first sensing channel **83** may produce a sensed event signal, such as an R-wave sensed event signal, that is passed to control circuit **80**. The R-wave sensed event signals may be used by control circuit **80** for determining

RRIs for detecting tachyarrhythmia and determining a need for therapy. An RRI is the time interval between two consecutively sensed R-waves and may be determined between consecutive R-wave sensed event signals received by control circuit **80** from sensing circuit **86**. For example, control circuit **80** may include a timing circuit **90** for determining RRIs between consecutive R-wave sensed event signals received from sensing circuit **86** and for controlling various timers and/or counters used to control the timing of therapy delivery by therapy delivery circuit **84**. Timing circuit **90** may additionally set time windows such as morphology template windows, morphology analysis windows or perform other timing related functions of ICD **14** including synchronizing cardioversion shocks or other therapies delivered by therapy delivery circuit **84** with sensed cardiac events.

(52) The R-wave sensed event signals may be used by control circuit **80** to trigger storage of a time segment of a cardiac electrical signal for processing and analysis for detecting noise in the cardiac electrical signal as described below. In some examples, sensing circuit **86** senses at least one cardiac electrical signal received by a sensing electrode vector selected from the available electrodes, e.g., electrodes **24**, **26**, **28**, **30** and housing **15**, for detecting R-waves and buffering multiple cardiac electrical signal segments, where each cardiac electrical signal segment corresponds to a detected R-wave, for processing and analysis for detecting noise. A single cardiac electrical signal sensed by first sensing channel **83** may be used for both R-wave sensing and analysis of cardiac electrical signal segments for noise detection and true, underlying tachyarrhythmia evidence determination. In other examples, R-waves are detected from the first cardiac electrical signal sensed by the first sensing channel **83** and segments of a second cardiac electrical signal sensed by the second sensing channel **85** may be buffered, with each segment corresponding to an R-wave sensed from the first cardiac electrical signal. Analysis for determining noise corruption may be performed on the second cardiac electrical signal segments. The second cardiac electrical signal may be received via a sensing electrode pair coupled to the second sensing channel **85** different than the sensing electrode pair coupled to the first sensing channel **83** for sensing than the first cardiac electrical signal and/or may be received by the same sensing electrode pair but processed differently, e.g., filtered differently, by the second sensing channel **85** to produce a second cardiac electrical signal sensed by sensing circuit **86** different than the first cardiac electrical signal.

(53) Memory **82** may be configured to store a predetermined number of cardiac electrical signal segments in a circulating buffer under the control of control circuit **80**, e.g., at least one, two, three or other number of cardiac electrical signal segments. Each segment may be written to memory **82** over a time interval extending before and after an R-wave sensed event signal produced by the first sensing channel **83**. Control circuit **80** may access stored cardiac electrical signal segments when confirmation of R-waves sensed by the first sensing channel **83** is required based on the detection of a predetermined number of tachyarrhythmia intervals, which may precede tachyarrhythmia detection. In some examples, R-waves sensed by the first sensing channel **83** may be rejected when a cardiac electrical signal sensed by the second sensing channel **85** is determined to be noise corrupted. In other examples, R-waves sensed by the first sensing channel **83** may be used by control circuit **80** to determine RRIs that may be counted as tachyarrhythmia intervals, but when a tachyarrhythmia detection criterion is satisfied based on the RRIs, control circuit **80** may withhold tachyarrhythmia detection if the second cardiac electrical signal is determined to be noise corrupted. Methods for determining that noise is present in a cardiac electrical signal are described below.

(54) Control circuit **80** is shown to include a tachyarrhythmia detector **92** configured to analyze signals received from sensing circuit **86** for detecting tachyarrhythmia. Tachyarrhythmia detector **92** may detect tachyarrhythmia based on R-wave sensed event signals meeting tachyarrhythmia criteria, such as a threshold number of sensed event signals occurring at a tachyarrhythmia interval. In some examples, a tachyarrhythmia detection based on the threshold number of sensed event

signals each occurring at a tachyarrhythmia interval may be rejected based on non-cardiac noise being detected using the techniques disclosed herein. Tachyarrhythmia detector **92** may be implemented in control circuit **80** as hardware, software and/or firmware that processes and analyzes signals received from sensing circuit **86** for detecting VT and/or VF. In some examples, the timing of R-wave sensed event signals received from sensing circuit **86** is used by timing circuit **90** to determine RRI between sensed event signals. These RRIs may also be referred to herein as “sensed event intervals.” Tachyarrhythmia detector **92** may include comparators and counters for counting RRIs determined by timing circuit **90** that fall into various rate detection zones for determining a ventricular rate or performing other rate- or interval-based assessment of R-wave sensed event signals for detecting and discriminating VT and VF.

(55) For example, tachyarrhythmia detector **92** may compare the RRIs determined by timing circuit **90** to one or more tachyarrhythmia detection interval zones, such as a tachycardia detection interval zone and a fibrillation detection interval zone. RRIs falling into a detection interval zone are counted by a respective VT interval counter or VF interval counter and in some cases in a combined VT/VF interval counter included in tachyarrhythmia detector **92**. The VF detection interval threshold may be set to 300 to 350 milliseconds (ms), as examples. For instance, if the VF detection interval is set to 320 ms, RRIs that are less than 320 ms are counted by the VF interval counter. When VT detection is enabled, the VT detection interval may be programmed to be in the range of 350 to 420 ms, or 400 ms as an example. In order to detect VT or VF, the respective VT or VF interval counter is required to reach a threshold “number of intervals to detect” (NID).

(56) As an example, the NID to detect VT may require that the VT interval counter reaches 32 VT intervals counted out of the most recent 32 consecutive RRIs. The NID required to detect VF may be programmed to 18 VF intervals out of the most recent 24 consecutive RRIs or 30 VF intervals out 40 consecutive RRIs, as examples. When a VT or VF interval counter reaches an NID threshold, a ventricular tachyarrhythmia may be detected by tachyarrhythmia detector **92**. The NID may be programmable and range from as low as 12 to as high as 40, with no limitation intended. VT or VF intervals may be detected consecutively or non-consecutively out of the specified number of most recent RRIs. In some cases, a combined VT/VF interval counter may count both VT and VF intervals and detect a tachyarrhythmia episode based on the fastest intervals detected when a specified NID is reached.

(57) Tachyarrhythmia detector **92** may be configured to perform other signal analysis for determining if other detection criteria are satisfied before detecting VT or VF, such as R-wave morphology criteria, onset criteria, and noise and oversensing rejection criteria. Examples of parameters that may be determined from cardiac electrical signals received from sensing circuit **86** for determining noise corruption that may cause withholding of a VT or VF detection are described below.

(58) To support these additional analyses, sensing circuit **86** may pass a digitized electrocardiogram (ECG) signal to control circuit **80** for morphology analysis performed by tachyarrhythmia detector **92**. A cardiac electrical signal from the selected sensing vector, e.g., from first sensing channel **83** and/or the second sensing channel **85**, may be passed through a filter and amplifier, provided to a multiplexer and thereafter converted to a multi-bit digital signal by an analog-to-digital converter, all included in sensing circuit **86**, for storage in memory **82**. Memory **82** may include one or more circulating buffers to temporarily store digital cardiac electrical signal segments for analysis performed by control circuit **80**. Control circuit **80** may be a microprocessor-based controller that employs digital signal analysis techniques to characterize the digitized signals stored in memory **82** to recognize and classify the patient's heart rhythm employing any of numerous signal processing methodologies for analyzing cardiac signals and cardiac event waveforms, e.g., R-waves. As described below, processing and analysis of digitized signals may include determining signal features or metrics for detecting noise present in the cardiac electrical signal(s). When noise is detected, a tachyarrhythmia detection based on RRIs may be withheld to inhibit a tachyarrhythmia

therapy. Alternatively, a tachyarrhythmia therapy may be withheld in response to a tachyarrhythmia detection made when noise is also detected.

(59) Therapy delivery circuit **84** includes charging circuitry, one or more charge storage devices such as one or more high voltage capacitors and/or low voltage capacitors, and switching circuitry that controls when the capacitor(s) are discharged across a selected pacing electrode vector or CV/DF shock vector. Charging of capacitors to a programmed pulse amplitude and discharging of the capacitors for a programmed pulse width may be performed by therapy delivery circuit **84** according to control signals received from control circuit **80**. Control circuit **80** may include various timers or counters that control when cardiac pacing pulses are delivered. For example, timing circuit **90** may include programmable digital counters set by a microprocessor of the control circuit **80** for controlling the basic pacing time intervals associated with various pacing modes or ATP sequences delivered by ICD **14**. The microprocessor of control circuit **80** may also set the amplitude, pulse width, polarity or other characteristics of the cardiac pacing pulses, which may be based on programmed values stored in memory **82**.

(60) In response to detecting VT or VF, control circuit **80** may schedule a therapy and control therapy delivery circuit **84** to generate and deliver the therapy, such as ATP and/or CV/DF therapy. Therapy can be generated by initiating charging of high voltage capacitors via a charging circuit, both included in therapy delivery circuit **84**. Charging is controlled by control circuit **80** which monitors the voltage on the high voltage capacitors, which is passed to control circuit **80** via a charging control line. When the voltage reaches a predetermined value set by control circuit **80**, a logic signal is generated on a capacitor full line and passed to therapy delivery circuit **84**, terminating charging. A CV/DF pulse is delivered to the heart under the control of the timing circuit **90** by an output circuit of therapy delivery circuit **84** via a control bus. The output circuit may include an output capacitor through which the charged high voltage capacitor is discharged via switching circuitry, e.g., an H-bridge, which determines the electrodes used for delivering the cardioversion or defibrillation pulse and the pulse wave shape.

(61) In some examples, the high voltage therapy circuit configured to deliver CV/DF shock pulses can be controlled by control circuit **80** to deliver pacing pulses, e.g., for delivering ATP, post shock pacing pulses or ventricular pacing pulses. In other examples, therapy delivery circuit **84** may include a low voltage therapy circuit for generating and delivering pacing pulses for a variety of pacing needs.

(62) It is recognized that aspects of the methods disclosed herein for determining noise corruption of a cardiac electrical signal may be implemented in a medical device that is used for monitoring cardiac electrical signals by sensing circuit **86** and control circuit **80** without having therapy delivery capabilities or in a pacemaker that monitors cardiac electrical signals and delivers cardiac pacing therapies by therapy delivery circuit **84**, without high voltage therapy capabilities, such as CV/DF shock capabilities.

(63) Control parameters utilized by control circuit **80** for sensing cardiac events and controlling therapy delivery may be programmed into memory **82** via telemetry circuit **88**. Telemetry circuit **88** includes a transceiver and antenna for communicating with external device **40** (shown in FIG. 1A) using RF communication or other communication protocols as described above. Under the control of control circuit **80**, telemetry circuit **88** may receive downlink telemetry from and send uplink telemetry to external device **40**.

(64) FIG. 4 is a diagram of circuitry included in sensing circuit **86** having first sensing channel **83** and second sensing channel **85** according to one example. First sensing channel **83** may be selectively coupled via switching circuitry included in sensing circuit **86** to a first sensing electrode vector including at least one electrode carried by extra-cardiovascular lead **16** for sensing a first cardiac electrical signal. In some examples, first sensing channel **83** may be coupled to a sensing electrode vector that is a short bipole, having a relatively shorter inter-electrode distance or spacing than the second electrode vector coupled to second sensing channel **85**. First sensing channel **83**

may be coupled to a sensing electrode vector that is approximately vertical (when the patient is in an upright position) or approximately aligned with the cardiac axis to increase the likelihood of a relatively high R-wave signal amplitude. In one example, the first sensing electrode vector may include pace/sense electrodes **28** and **30**. In other examples, the first sensing electrode vector coupled to sensing channel **83** may include a defibrillation electrode **24** and/or **26**, e.g., a sensing electrode vector between pace/sense electrode **28** and defibrillation electrode **24** or between pace/sense electrode **30** and either of defibrillation electrodes **24** or **26**. In still other examples, the first sensing electrode vector may be between defibrillation electrodes **24** and **26**.

(65) Sensing circuit **86** includes second sensing channel **85** for sensing a second cardiac electrical signal in some examples. For instance, second sensing channel **85** may receive a raw cardiac electrical signal from a second sensing electrode vector, for example from a vector that includes one electrode **24**, **26**, **28** or **30** carried by lead **16** paired with housing **15**. Second sensing channel **85** may be selectively coupled to other sensing electrode vectors, which may form a relatively longer bipole having an inter-electrode distance or spacing that is greater than the sensing electrode vector coupled to first sensing channel **83** in some examples. The second sensing electrode vector may be, but not necessarily, approximately orthogonal to the first channel sensing electrode vector in some cases. For instance, defibrillation electrode **26** and housing **15** may be coupled to second sensing channel **85** to provide the second cardiac electrical signal. As described below, the second cardiac electrical signal received by second sensing channel **85** via a long bipole may be used by control circuit **80** for analysis and detection of noise. The long bipole coupled to second sensing channel **85** may provide a relatively far-field or more global cardiac signal compared to the relatively shorter bipole coupled to the first sensing channel. In other examples, any vector selected from the available electrodes, e.g., electrodes **24**, **26**, **28**, **30** and/or housing **15**, may be included in a sensing electrode vector coupled to second sensing channel **85**. The sensing electrode vectors coupled to first sensing channel **83** and second sensing channel **85** may be different sensing electrode vectors, which may have no common electrodes or only one common electrode but not both in some examples.

(66) In other examples, however, the sensing electrode vectors coupled to the first sensing channel **83** and the second sensing channel **85** may be the same sensing electrode vector. The two sensing channels **83** and **85** may include different filters, amplifiers, or other signal processing circuitry such that two different signals are sensed by the respective sensing channels **83** and **85** and different analyses may be performed on the two signals. For example, the first sensing channel **83** may sense a first cardiac electrical signal by filtering and processing the received cardiac electrical signal for detecting R-waves in response to an R-wave sensing threshold crossing for determining RRTs. The second sensing channel **85** may sense a second cardiac electrical signal different than the first by filtering and processing the received cardiac electrical signal for passing signal segments to control circuit **80** for analysis for noise detection. The first sensing channel **83** may apply relatively narrower band pass filtering, and the second sensing channel **85** may apply relatively wider band pass filtering and, in some examples, notch filtering to provide two different sensed cardiac electrical signals, received via the same sensing electrode vector.

(67) In the illustrative example shown in FIG. **4**, the electrical signals developed across the first sensing electrode vector, e.g., electrodes **28** and **30**, are received by first sensing channel **83** and electrical signals developed across the second sensing electrode vector, e.g., electrodes **26** and housing **15**, are received by second sensing channel **85**. The cardiac electrical signals are provided as differential input signals to the pre-filter and pre-amplifier **62** or **72**, respectively, of first sensing channel **83** and second sensing channel **85**. Non-physiological high frequency and DC signals may be filtered by a low pass or bandpass filter included in each of pre-filter and pre-amplifiers **62** and **72**, and high voltage signals may be removed by protection diodes included in pre-filter and pre-amplifiers **62** and **72**. Pre-filter and pre-amplifiers **62** and **72** may amplify the pre-filtered signal by a gain of between 10 and 100, and in one example a gain of 17.5, and may convert the differential

signal to a single-ended output signal passed to analog-to-digital converter (ADC) **63** in first sensing channel **83** and to ADC **73** in second sensing channel **85**. Pre-filters and amplifiers **62** and **72** may provide anti-alias filtering and noise reduction prior to digitization.

(68) ADC **63** and ADC **73**, respectively, convert the first cardiac electrical signal from an analog signal to a first digital bit stream and the second cardiac electrical signal to a second digital bit stream. In one example, ADC **63** and ADC **73** may be sigma-delta converters (SDC), but other types of ADCs may be used. In some examples, the outputs of ADC **63** and ADC **73** may be provided to decimators (not shown), which function as digital low-pass filters that increase the resolution and reduce the sampling rate of the respective first and second cardiac electrical signals.

(69) The digital outputs of ADC **63** and ADC **73** may each be passed to respective filters **64** and **74**, which may be digital bandpass filters. The bandpass filters **64** and **74** may have the same or different bandpass frequencies. For example, filter **64** may have a relatively narrow bandpass of approximately 13 Hz to 39 Hz, as an example, for passing cardiac electrical signals such as R-waves typically occurring in this frequency range. Filter **74** of the second sensing channel **85** may have a relatively wider bandpass of approximately 2.5 to 100 Hz as an example. In some examples, second sensing channel **85** may further include a notch filter **76** to filter 60 Hz and/or 50 Hz noise signals and in some cases harmonics of 60 Hz and/or 50 Hz, e.g., 120 Hz and/or 100 Hz.

(70) The narrow bandpass filtered signal in first sensing channel **83** is passed from filter **64** to rectifier **65** to produce a filtered, rectified signal. First sensing channel **83** includes an R-wave detector **66** for sensing cardiac events in response to the first cardiac electrical signal crossing an R-wave sensing threshold. R-wave detector **66** may include an auto-adjusting sense amplifier, comparator and/or other detection circuitry that compares the filtered and rectified cardiac electrical signal to an R-wave sensing threshold in real time and produces an R-wave sensed event signal **68** when the cardiac electrical signal crosses the R-wave sensing threshold outside of a post-sense blanking interval. The R-wave sensing threshold may be a multi-level sensing threshold, e.g., as disclosed in commonly assigned U.S. Pat. No. 10,252,071 (Cao, et al.), incorporated herein by reference in its entirety. Briefly, the multi-level sensing threshold may have a starting sensing threshold value held for a time interval, which may be equal to a tachycardia detection interval or expected R-wave to T-wave interval, then drops to a second sensing threshold value held until a drop time interval expires, which may be 1 to 2 seconds long. The sensing threshold drops to a minimum sensing threshold, which may correspond to a programmed sensitivity sometimes referred to as the “sensing floor,” after the drop time interval. In other examples, the R-wave sensing threshold used by R-wave detector **66** may be set to a starting value based on the peak amplitude determined during the most recent post-sense blanking interval and decay linearly or exponentially over time until reaching a minimum sensing threshold. The techniques described herein are not limited to a specific behavior of the sensing threshold or specific R-wave sensing techniques. Instead, one or more decay rates and intervals and/or one or more step-wise adjustments may be utilized to adjust the R-wave sensing threshold.

(71) The wideband filtered digital cardiac electrical signal **78** from second sensing channel **85** may be passed to memory **82** for buffering a segment of the second cardiac electrical signal **78** in response to an R-wave sensed event signal **68** produced by the first sensing channel **83**. The wideband filtered signal may be notch filtered prior to being buffered in memory **82**. In some examples, output from the wideband filter **74** and output from the notch filter **76** are buffered to provide a pre-notch filtered signal and a post-notch filtered signal for use in the noise determination analyses described below. In some examples, the buffered segment of the second cardiac electrical signal **78** is rectified by rectifier **75** before being stored in memory **82**. In some cases, both the filtered, non-rectified signal **78** and the rectified signal **79** are passed to control circuit **80** and/or memory **82** for use in determining features of the second cardiac electrical signal. A feature of the second cardiac electrical signal may be determined from a second cardiac electrical signal segment where each segment extends over a time interval that encompasses the time point of an R-wave

sensed event signal produced by the first sensing channel **83**.

(72) Control circuit **80** is configured to detect tachyarrhythmia based on cardiac events detected from at least one cardiac electrical signal sensed by sensing circuit **86**. For example, control circuit **80** may be configured to detect tachyarrhythmia when a threshold number of sensed cardiac events each occur at a tachyarrhythmia interval. Control circuit **80** may buffer segments of a sensed cardiac electrical signal in memory **82** and retrieve stored signal segments from memory **82** for analysis when a lower threshold number of tachyarrhythmia intervals have been detected, before the tachyarrhythmia detection threshold is reached. In some examples, RRI for detecting tachyarrhythmia intervals are determined from the first cardiac electrical signal sensed by first sensing channel **83**, and cardiac electrical signal segments are buffered from the second cardiac electrical signal received by control circuit **80** from second sensing channel **85** for noise analysis when the lower threshold number of tachyarrhythmia intervals is detected. Analysis of the second cardiac electrical signal segments may be performed for use in detecting non-cardiac noise before the detection threshold number of tachyarrhythmia intervals (NID) is reached, as described below. In other examples, a single cardiac electrical signal sensed by sensing circuit **86** is used for sensing R-waves for determining RRI and counting tachyarrhythmia intervals and buffering cardiac electrical signal segments for use in determining noise corruption, where each buffered segment may be associated with one sensed R-wave.

(73) The configuration of sensing channels **83** and **85** as shown in FIG. **4** is illustrative in nature and should not be considered limiting of the techniques described herein. The sensing channels **83** and **85** of sensing circuit **86** may include more or fewer components than illustrated and described in FIG. **4** and some components may be shared between sensing channels **83** and **85**. For example, one or more of pre-filter and pre-amplifiers **62/72** and/or ADC **63/73** may be shared components between sensing channels **83** and **85** with a single, sensed signal output split to two sensing channels for subsequent processing and analysis. Sensing circuit **86** and control circuit **80** include circuitry configured to perform the functionality attributed to ICD **14** in detecting suspected noise and rejecting or withholding tachyarrhythmia detection in response to determining noise corruption as disclosed herein.

(74) FIG. **5** is a flow chart **100** of a method for determining that a cardiac electrical signal is corrupted by noise according to one example. At block **102**, sensing circuit **86** senses a cardiac event signal, e.g., an R-wave. Control circuit **80** may determine a maximum amplitude associated with the sensed cardiac event at block **104**. In some examples, the maximum amplitude may be determined from the same cardiac electrical signal from which the cardiac event was sensed. In other examples, control circuit **80** may buffer a segment of a different cardiac electrical signal, e.g., the wide band and notch-filtered signal from the second sensing channel **85**, in memory **82**. The maximum amplitude may be a maximum peak to peak amplitude determined from a segment of the cardiac electrical signal that is buffered over a time interval that includes the time of the cardiac sensed event signal from the first sensing channel **83**, e.g., R-wave sensed event signal **68**. In other examples, a maximum peak amplitude of a wide band and notch filtered rectified signal may be determined at block **104**. The maximum amplitude may be stored in an amplitude buffer in memory **82** that stores a predetermined number of maximum amplitudes, each associated with a sensed cardiac event, in a first in first out basis. Each maximum amplitude may be stored in the amplitude buffer with a timestamp corresponding to the time of the maximum peak or corresponding to the time of the associated sensed event signal. In some instances, a very large signal may saturate the ADC of the sensing channel. In this case, the maximum amplitude may be flagged as invalid in the amplitude buffer. When the signal is clipped, exceeding the ADC input range, the maximum peak to peak amplitude of the input range may be stored as the maximum amplitude but may be labeled as invalid or ignored by control circuit **80** for determining when true tachyarrhythmia evidence criteria are met at block **114** as described below. An over-range amplitude may occur when a large noise signal occurs or when the cardiac electrical signal is erratic.

(75) At block **106**, control circuit **80** may determine if a required number of events have been sensed for determining if suspected noise criteria are met at block **108**. For example, six to eighteen maximum peak amplitudes may be stored in amplitude buffer in memory **82**. In some examples, three to fifteen, e.g., eleven to thirteen, maximum peak amplitudes are stored in the amplitude buffer in memory **82**. If a minimum number of maximum amplitudes are not stored in the buffer, the process returns to block **102** until the required number of maximum amplitudes are determined and stored in the amplitude buffer. A minimum number, e.g., at least three maximum amplitudes, may be required in order to determine if suspected noise criteria are met. In some examples, the buffer may be required to be filled. In other examples at least three out of thirteen buffer locations, or another predetermined number of maximum amplitudes, may be required for proceeding to determine if suspected noise criteria are met.

(76) When at least the required number of maximum amplitudes is stored, control circuit **80** may determine if the stored maximum amplitudes meet suspected noise criteria at block **108**. Control circuit **80** may apply one or more requirements to the maximum amplitudes stored in the amplitude buffer to determine if suspected noise criteria are met at block **108**. In one example, control circuit **80** determines if at least one of the stored peak to peak amplitudes is less than a noise amplitude threshold. In other examples, control circuit **80** may identify two or more of the lowest maximum amplitudes and determine a mean or median of selected number of lowest maximum amplitudes. For instance, control circuit **80** may determine a mean lowest maximum amplitude as the average of the two smallest maximum amplitude values stored in the maximum amplitude buffer. This mean lowest maximum amplitude may be compared to the noise amplitude threshold. The noise amplitude threshold may be set between 0.5 milliVolts (mV) and 2.5 mV, as examples. In one example, the noise amplitude threshold is set to 2.0 mV. When the mean lowest maximum amplitude is less than 2.0 mV (or another selected noise amplitude threshold), the suspected noise criteria may be met at block **108**.

(77) In some examples, the noise amplitude threshold may be set based on the peak amplitude of known cardiac events, e.g., based on the peak amplitude of one or more R-waves sensed during a normal, resting heart rate. For instance, a median R-wave amplitude may be determined and the noise amplitude threshold may be set to a fraction of the median R-wave amplitude. The noise amplitude threshold may be set to one-fourth, one fifth, one-sixth or other portion of the sensed R-wave amplitude, as examples.

(78) In other examples, control circuit **80** may determine a ratio of the lowest maximum amplitude to the greatest maximum amplitude from the stored maximum amplitudes. Control circuit **80** may compare the ratio to a threshold ratio. The lowest maximum amplitude, which may be a mean or median lowest maximum amplitude, is determined from one or more of the smallest maximum amplitudes stored in the amplitude buffer. The greatest maximum amplitude, which may be a mean or median greatest maximum amplitude, may be determined from one or more of the highest maximum amplitudes stored in the amplitude buffer. In one example, the ratio of the mean of the two lowest maximum amplitudes to the one greatest maximum amplitude stored in the amplitude buffer is determined and compared to a ratio threshold. The ratio threshold may be set to $2/5$, $1/5$, $1/6$, $1/7$, $1/8$ or other selected ratio. In one example, the ratio threshold is $1/6$. When the ratio of the mean lowest maximum amplitude (e.g., determined as a mean from the smallest two maximum amplitudes) to the single greatest maximum amplitude is less than $1/6$, suspected noise criteria are met at block **108**.

(79) In some examples, the lowest maximum amplitude (which may be determined as a mean of two or more of the smallest maximum amplitudes stored in the amplitude buffer) may be required to be less than a noise threshold amplitude, e.g., less than 2.0 mV, and the ratio of the lowest maximum amplitude to the greatest maximum amplitude may be required to be less than a ratio threshold, e.g., less than $1/6$, for control circuit **80** to determine that suspected noise criteria are met at block **108**. The suspected noise criteria generally require that at least one of the stored maximum

amplitudes is relatively small, e.g., less than the noise amplitude threshold, as evidence of relatively lower amplitude noise signals which may be present in the wide band and notch filtered signal and that the ratio of the lowest maximum amplitude to the highest maximum amplitude is relatively small, as evidence of at least one relatively large signal which may correspond to a true R-wave or fibrillation wave occurring during noise corruption. As indicated above, a maximum peak to peak amplitude in the amplitude buffer that is equal to the ADC input range (indicating a saturated or clipped signal) may be ignored for the purposes of determining a greatest maximum amplitude used in applying suspected noise criteria in some examples. In other examples, the maximum peak to peak amplitude corresponding to the ADC input range may be used as the highest maximum amplitude when comparing the ratio threshold to the ratio of the lowest maximum amplitude to the highest maximum amplitude.

(80) As another example, applying the suspected noise criteria by control circuit **80** at block **108** may include determining the maximum peak to peak amplitude from the wide band filtered signal segment prior to notch filtering (e.g., output of filter **74** in FIG. **4**) and determining the maximum peak to peak amplitude from the wide band and notch filtered signal (e.g., from signal **78** in FIG. **4**). Control circuit **80** may determine the ratio of the post-notch filtered maximum peak to peak amplitude to the pre-notch filtered peak to peak amplitude and compare the ratio to a relative noise threshold. The notch filter **76** is expected to attenuate 50-60 Hz noise. Accordingly, the maximum peak to peak amplitude of the notch filtered signal is expected to be significantly lower than the maximum peak to peak amplitude of the wide band filtered signal (output of filter **74**) when the R-wave sensed event signal that triggers buffering of the signal segment from the second channel **85** is an oversensed noise signal.

(81) Therefore, when the maximum peak to peak amplitude of the notch filtered signal is associated with a noise signal, the ratio is expected to be low, less than a relative noise threshold ratio. When the maximum peak to peak amplitude is associated with a true cardiac event signal, the true cardiac event signal is expected to be minimally attenuated by notch filtering. In this case, the ratio of the post-notch filtered maximum peak to peak amplitude to the pre-notch filtered maximum peak to peak amplitude is expected to be close to one. Control circuit **80** may set the relative noise threshold between 0.5 and 0.9 or between 0.7 and 0.8 as examples. When the determined ratio associated with at least one sensed cardiac event is less than the relative noise threshold, suspected noise criteria may be met at block **18**. When the ratio is greater than the relative noise threshold, the suspected noise criteria may be unmet. In some examples, control circuit **80** may compare the relative noise threshold to the ratio of the post- to pre-notch filtered maximum amplitudes determined for the lowest maximum peak to peak amplitude stored in the amplitude buffer (determined from the post-notch filtered signal). In other examples, control circuit **80** may compare the relative noise threshold to one or more post- to pre-notch filtered maximum amplitude ratios corresponding to maximum peak to peak amplitudes stored in the amplitude buffer associated with sensed cardiac event signals sensed at or less than a tachyarrhythmia interval. In still other examples, control circuit **80** may compare the relative noise threshold to each post- to pre-notch filtered maximum amplitude ratio that corresponds to maximum peak to peak amplitude stored in the amplitude buffer that is less than a noise threshold amplitude.

(82) The suspected noise criteria applied by control circuit **80** at block **108** may be met when at least one maximum peak to peak amplitude is less than the noise threshold amplitude, a ratio of the lowest maximum peak to peak amplitude to the greatest maximum peak to peak amplitude is less than a ratio threshold, and/or a post- to pre-notch filtered maximum amplitude ratio is less than a relative noise threshold or any combination thereof. In various examples, the suspected noise criteria may include one, two, three or more requirements applied to the maximum amplitudes stored in the amplitude buffer. The suspected noise criteria may be met when at least one requirement is met, two out of three requirements are met, or when all specified requirements are met, e.g., two out of two requirements.

(83) When control circuit **80** determines that suspected noise criteria are not met at block **108**, control circuit **80** adjusts a counter used to determine how many times suspected noise criteria are met over the most recent N sensed cardiac events. For example, control circuit **80** may move a zero into a first in first out suspected noise buffer in memory **82** configured to store a zero each time suspected noise criteria are not met and a one each time suspected noise criteria are met. The buffer may store the same number of values as the maximum amplitude buffer in some examples but may store a different number of values.

(84) When control circuit **80** determines that the suspected noise criteria are met at block **108**, before counting the suspected noise determination by moving a one into the suspected noise buffer, control circuit **80** may verify that true tachyarrhythmia evidence criteria are unmet at block **114**. Control circuit **80** may apply true tachyarrhythmia evidence criteria to the maximum amplitude data stored in the amplitude buffer and/or to cardiac signal segments buffered in response to the sensed cardiac events. The true tachyarrhythmia evidence criteria may include one or more interval related requirements applied to time intervals determined between maximum amplitudes stored in the buffer (e.g., based on timestamps or based on the cardiac event intervals, e.g., RRs, associated with the corresponding sensed event signals, e.g., R-wave sensed event signals). The true tachyarrhythmia evidence criteria may additionally or alternatively include morphology criteria applied to at least one cardiac signal segment buffered in memory and associated with a sensed event signal. The tachyarrhythmia evidence criteria may be applied to determine when evidence of at least one true tachyarrhythmia interval (or true cardiac event interval) is determined and/or evidence of at least one true tachyarrhythmia morphology waveform is determined over the N sensed cardiac events associated with the maximum amplitudes buffered in the amplitude buffer. Techniques for determining true tachyarrhythmia evidence criteria are met at block **114** are described below, e.g., in conjunction with FIGS. 7-10. As indicated above, when a maximum peak to peak amplitude equal to the ADC input range is stored in the amplitude buffer, this over-range amplitude may be ignored by control circuit **80** for the purposes of determining if tachyarrhythmia evidence criteria are met since an over-range signal is unlikely to be a true cardiac event signal.

(85) When control circuit **80** determines that true tachyarrhythmia evidence criteria is met at block **114**, control circuit **80** may move a zero into the suspected noise buffer at block **110**. Even though the suspected noise criteria were met at block **108**, evidence of a true tachyarrhythmia event prevents control circuit **80** from counting the suspected noise determination toward a determination of noise corruption in the cardiac electrical signal used to withhold a tachyarrhythmia detection. In this way, control circuit **80** inhibits determining noise corruption that would withhold a tachyarrhythmia detection when other cardiac signal evidence suggests that the true underlying rhythm may be tachyarrhythmia.

(86) When control circuit **80** determines that the true tachyarrhythmia evidence criteria are not met at block **114**, control circuit **80** may count the suspected noise determination at block **116**, e.g., by setting a suspected noise flag. For instance, control circuit **80** may move a one into a first in first out suspected noise buffer to facilitate counting suspected noise determinations. The size of the suspected noise buffer configured to store a suspected noise determination as a 1 or a 0 may be between 6 and 18 and is 12 in one example. At block **118**, control circuit **80** may compare the suspected noise determination count to a noise corruption threshold. For example, control circuit **80** may count the number of ones stored in the suspected noise buffer. When the suspected noise count is greater than or equal to a noise corruption threshold, control circuit **80** may determine noise corruption of the cardiac electrical signal **120**. For instance, control circuit **80** may determine noise corruption of the cardiac electrical signal at block **120** in response to at least three suspected noise determinations over of the most recent twelve sensed cardiac events. It is noted that a suspected noise signal may not correspond to the currently sensed cardiac event. Rather, one or more of the maximum amplitudes stored in the amplitude buffer may correspond to a noise signal, which may be any of the most recent sensed events corresponding to the stored maximum amplitudes, when

suspected noise criteria are met.

(87) In response to determining noise corruption control circuit **80** may withhold making a rhythm detection or heart rate determination at block **122**. In the examples described herein, control circuit **80** may withhold detecting a tachyarrhythmia based on other tachyarrhythmia detection criteria being met when a noise corruption determination is made.

(88) FIG. **6** is diagram **200** of a first cardiac electrical signal **202** and a second cardiac electrical signal **232** that may be used for sensing cardiac events and determining noise corruption according to one example. In the first cardiac electrical signal **202**, two true R-waves **204** and **206** are sensed by the first sensing channel **83** resulting in respective R-wave sensed event signals **208** and **210** (labeled “VS”). However, due to non-cardiac noise present in the first cardiac electrical signal **202**, two noise signals **212** and **214** are each sensed by R-wave detector **66** resulting in false R-wave sensed event signals **216** and **218**. Each of the R-wave sensed event signals **208**, **210**, **216** and **218**, denoted by “VS” to indicate a ventricular sensed event, trigger buffering of a segment of the second cardiac electrical signal **232** over a respective time interval **233**, **235**, **241**, and **243**.

(89) Each time interval **233**, **235**, **241**, and **243** may extend before and after the time of the respective R-wave sensed event signal **208**, **210**, **216**, and **218** such that the associated buffered second cardiac electrical signal segment includes sample points before and after the time of the associated R-wave sensed event signal. Each time interval **233**, **235**, **241** and **243** may be about 360 ms in duration so that the respective second cardiac electrical signal segment buffered over the time interval includes 92 sample points when the sampling rate is 256 Hz with **24** of the sample points occurring after the R-wave sensed event signal that triggered the storage of the signal segment and 68 sample points extending from the R-wave sensed event signal earlier in time from the R-wave sensed event signal. Each cardiac electrical signal segment may be longer or shorter than 360 ms, but each cardiac electrical signal segment buffered over the respective time interval **233**, **235**, **241** or **243** encompasses the time of one, single R-wave sensed event signal. If an R-wave sensed event signal occurs before the end of the buffering time interval for an immediately preceding R-wave sensed event signal, control circuit **80** may skip buffering a time segment for that R-wave sensed event signal.

(90) The buffered signal segments may be notch filtered prior to determining maximum peak to peak amplitude data from the signal segments in some examples. Control circuit **80** determines a maximum peak to peak amplitude **234** of the buffered cardiac electrical signal segment from the sample points within time interval **233**. Maximum peak to peak amplitude **242** is determined from the buffered cardiac electrical signal segment within time interval **241**, and maximum peak to peak amplitudes **244** and **236** are determined from the cardiac electrical signal segments buffered within respective time intervals **243** and **235**. Control circuit **80** may determine each maximum peak to peak amplitude **234**, **236**, **242** and **244** by determining the difference between a maximum sample point amplitude and a minimum sample point amplitude determined from the second cardiac electrical signal sample points within each respective time interval **233**, **235**, **241** or **243**. In some examples, the maximum peak to peak amplitudes **234**, **236**, **241** and **244** may be determined from sample points occurring within a portion of the respective time interval **233**, **235**, **241** or **243**. For example, each time interval may be at least 300 ms long, e.g., 360 ms or longer. The maximum peak to peak amplitude **234** may be determined as the difference between the maximum sample point and the minimum sample point occurring within a shorter time interval **250** within time interval **233**. Time interval **250** may be 200 ms or less in some examples. For instance, when time interval **233** is 360 ms, time interval **250** may be 188 ms and includes the time point of R-wave sensed event signal **208**. For a 256 Hz sampling rate, the 188 ms time interval may include 48 sample points centered on the time point of the R-wave sensed event signal **208**. Likewise, maximum peak to peak amplitudes **236**, **242** and **244** may be determined from sample points occurring over a shortened portion of respective time intervals **235**, **241** and **243**.

(91) These maximum peak to peak amplitudes **234**, **242**, **244** and **236** may be stored in a first-in-

first-of amplitude buffer in memory **82** so that the maximum peak to peak amplitude of each one of multiple second cardiac electrical signal segments, associated with each R-wave sensed event signal **208**, **210**, **216** and **218**, is buffered in a rolling, first-in-first-out amplitude buffer. The amplitude buffer may store a predetermined number of maximum amplitudes, e.g., 3 to 20 maximum amplitudes corresponding to the most recent 3 to 20 cardiac electrical signal segments and associated R-wave sensed event signals. In one example, at least 10 to 13 maximum amplitudes are stored.

(92) Each of the maximum peak to peak amplitudes **234**, **236**, **242** and **244** may be stored with a timestamp or an associated cardiac event interval, e.g., RRI, determined by timing circuit **90** for the associated R-wave sensed event signal **208**, **210**, **216** or **218** (from the most recent preceding R-wave sensed event signal). Maximum peak to peak amplitude **242** may be stored with the RRI **219**, which is the time interval between the R-wave sensed event signal **216** associated with maximum peak to peak amplitude **242** and the immediately preceding R-wave sensed event signal **208**. Maximum peak to peak amplitude **244** may be stored with RRI **220**, and maximum peak to peak amplitude **236** may be stored with RRI **222**, for example.

(93) Control circuit **80** may apply suspected noise criteria to the buffered maximum peak to peak amplitudes **234**, **236**, **242** and **244**, e.g., as described above in conjunction with FIG. 5 at block **108**. When the suspected noise criteria are met, control circuit **80** may analyze the morphology of at least one second cardiac electrical signal segment stored over a respective time interval, which may be all or a portion of the respective time interval **233**, **235**, **241** or **243**, to verify that a tachyarrhythmia morphology is not present before determining suspected noise and setting a suspected noise flag. For example, control circuit **80** may identify the greatest maximum peak to peak amplitude that is stored in the maximum amplitude buffer. For instance, peak to peak amplitude **234** in the example shown in FIG. 6 may be the greatest maximum amplitude stored. Control circuit **80** may determine one or more gross morphology metrics as described below in conjunction with FIGS. 9 and 10 based on at least a signal segment corresponding to the greatest maximum peak to peak amplitude **234** in the amplitude buffer. Control circuit **80** may be configured to determine if the morphology of the cardiac signal segment associated with the greatest maximum amplitude meets true tachyarrhythmia evidence criteria (e.g., at block **114** of FIG. 5). If so, control circuit **80** may not determine suspected noise and, in response, move a zero into the suspected noise buffer. If control circuit **80** determines that the gross morphology metric does not meet true tachyarrhythmia evidence criteria, control circuit **80** may determine suspected noise and move a one into the suspected noise buffer. As described below, the gross morphology metric may be determined using all sample points spanning the time interval **233** over which the cardiac electrical signal segment is buffered. In some examples, a gross morphology amplitude metric and a gross morphology signal width metric may be determined. When control circuit **80** determines that at least one gross morphology metric does not meet the true tachyarrhythmia evidence criteria, control circuit **80** may determine suspected noise based on the analysis of the maximum peak to peak amplitudes stored in the amplitude buffer.

(94) In some examples, control circuit **80** may analyze the timestamps or associated RRIs **219**, **220** **222** that may be stored with each maximum peak to peak amplitude. As described below in conjunction with FIG. 7, control circuit **80** may analyze intervals between the sensed events associated with the maximum peak to peak amplitudes for determining if interval-based criteria of true tachyarrhythmia evidence are met at block **114** of FIG. 5. For example, control circuit **80** may identify buffered maximum amplitudes **234** and **236** that are greater than a threshold amplitude **252** as suspected true cardiac events. As described below in conjunction with FIG. 7, the threshold amplitude **252** may be set to percentage or fraction of the greatest maximum amplitude **234**. In one example, control circuit **80** sets the threshold amplitude **252** to five-eighths of the greatest maximum amplitude **234**. Using the timestamps or associated RRIs stored with each of the buffered maximum amplitudes, control circuit **80** may determine a time interval **254** between two

consecutive maximum peak to peak amplitudes that are greater than the threshold amplitude 252. This time interval 254 may be determined as a suspected true cardiac event interval. When this time interval falls within a tachyarrhythmia interval range, control circuit 80 may determine evidence of a true tachyarrhythmia.

(95) FIG. 7 is a flow chart 300 of a method for determining when tachyarrhythmia evidence criteria are not met so that a determination of suspected noise can be made based on buffered maximum amplitudes according to one example. At block 301, control circuit 80 moves a new maximum peak to peak amplitude into the amplitude buffer. Control circuit 80 may determine the new maximum peak to peak amplitude from a portion of a cardiac electrical signal segment buffered in response to a cardiac sensed event signal as described above in conjunction with FIG. 5.

(96) At block 302, control circuit 80 identifies a greatest maximum amplitude from among the buffered maximum amplitudes in the amplitude buffer. At block 304, control circuit 80 determines a threshold amplitude, which may be based on the buffered maximum amplitudes. The threshold amplitude, e.g., threshold amplitude 252 in FIG. 6, may be based on the greatest maximum amplitude identified at block 302. For example, control circuit 80 may identify the greatest maximum amplitude and set the threshold amplitude as a percentage or fraction of the greatest maximum amplitude. For instance, the threshold amplitude may be set to 50% to 80% of the greatest maximum amplitude. In one example, control circuit 80 sets the threshold amplitude to five-eighths of the greatest maximum amplitude. In other examples, control circuit 80 may identify two or more of the highest maximum amplitudes in the amplitude buffer and determine the threshold amplitude at block 304 based on two or more of the highest maximum amplitudes. For example, the threshold amplitude may be set to a percentage or fraction of a mean, median, or nth highest maximum amplitude determined from two or more of the highest maximum amplitudes. The threshold amplitude may alternatively be a predetermined, programmed value.

(97) The threshold amplitude is set to discriminate between signals that are suspected true cardiac event signals and suspected noise signals based on amplitude. At block 306, control circuit 80 compares each buffered maximum amplitude value to the threshold amplitude to identify suspected true cardiac events. Each buffered maximum amplitude value that is greater than the threshold amplitude is identified as a suspected true cardiac event, e.g., a true R-wave. Based on the timestamp or sensed event interval stored with each of the identified suspected true cardiac events, control circuit 80 may determine event intervals associated with the suspected true cardiac events at block 308. For example, control circuit 80 may determine the time interval between two maximum amplitudes identified as consecutive suspected true cardiac events. The time interval may be determined by control circuit 80 as the difference between the timestamps of two maximum amplitudes that are both greater than the threshold amplitude and are therefore consecutive suspected true cardiac events. The time interval may alternatively be determined based on cardiac event intervals, e.g., RRIs, determined by timing circuit 90 between consecutive sensed event signals. For instance, control circuit 80 may determine the suspected event interval by summing all intervening RRIs between two consecutive sensed event signals associated with maximum amplitudes greater than the threshold amplitude. Control circuit 80 may determine the suspected cardiac event interval by summing two or more consecutive RRIs that occur between two maximum amplitudes that are identified as suspected true cardiac events.

(98) With reference to FIG. 6, as an example, control circuit 80 may determine that the maximum peak to peak amplitudes 234 and 236 are greater than the threshold amplitude and maximum peak to peak amplitudes 242 and 244 are less than the threshold amplitude. As a result, control circuit 80 may determine a suspected true cardiac event interval as the time interval from VS signal 208 to VS signal 210. Control circuit 80 may sum the RRIs 219, 220 and 222 occurring between the VS event signals 208 and 210 that are associated with two consecutive maximum peak to peak amplitudes that are greater than the threshold amplitude 252. As noted above, in other examples control circuit 80 may determine the difference between a timestamp stored for maximum peak to

peak amplitude **234** and maximum peak to peak amplitude **236** as the suspected true cardiac event interval.

(99) Returning to FIG. 7, in some instances, a single suspected true cardiac event interval may be determined at block **308**. In other instances, two or more suspected true cardiac event intervals may be determined, or no suspected true cardiac event intervals may be determined by control circuit **80** based on the buffered maximum amplitudes. The number of suspected true cardiac event intervals determined at block **308** depends on the number of suspected cardiac events identified from the buffered maximum amplitudes at block **306**. When the threshold amplitude is set based on the greatest maximum amplitude in the amplitude buffer, at least the one greatest maximum amplitude will be identified as being associated with a suspected true cardiac event. A single suspected true cardiac event identified at block **306** may yield no determinable value of a suspected true cardiac event interval from the buffered data since two maximum amplitudes greater than the threshold amplitude are not available for determining a value of the suspected true cardiac event interval.

(100) If only one suspected true cardiac event is identified at block **306**, the event may be a noise spike or a true R-wave or fibrillation wave. The remaining buffered maximum amplitudes may be associated with noise signals or low amplitude fibrillation waves. A true cardiac event interval associated with the one suspected true cardiac event may extend earlier than the timestamp of the oldest buffered maximum amplitude. The suspected true cardiac event interval value, however, is indeterminable. It may be assumed to be greater than the time interval to the oldest buffered maximum amplitude or the single event having the greatest maximum amplitude may be a noise signal. In this case, control circuit **80** may determine that true tachyarrhythmia interval criteria are met at block **312** in order to advance to block **316** to perform morphology analysis. Morphology analysis may be performed for detecting true tachyarrhythmia evidence when a suspected true cardiac event interval is indeterminable due to a single suspected true cardiac event being identified.

(101) Control circuit **80** may determine one single suspected true cardiac event interval at block **308** when only two suspected true cardiac events are identified at block **306**. When three or more suspected true cardiac events are identified at block **304**, control circuit **80** may determine two or more suspected true cardiac event intervals ($n-1$ intervals when n suspected true cardiac events are identified). When at least one suspected true cardiac event interval is determined at block **308**, control circuit **80** may compare the suspected true cardiac event interval(s) to true tachyarrhythmia interval evidence criteria at block **312**. In one example, if at least one suspected true cardiac event interval falls within a true tachyarrhythmia evidence interval range, the true tachyarrhythmia interval evidence criteria may be determined to be met by control circuit **80**.

(102) The true tachyarrhythmia evidence interval range may be based on a programmed tachyarrhythmia detection interval or set to a predetermined range in various examples. For instance, when VT detection is enabled, the true tachyarrhythmia evidence interval range may have an upper limit set based on the VT detection interval, e.g., the VT detection interval plus an offset. If VT detection is disabled, the true tachyarrhythmia evidence interval range may have an upper limit set based on the VF detection interval, e.g., the VF detection interval plus an offset. As described above, a programmed VF detection interval may be 320 ms as an example. A programmed VT detection interval may be 400 ms as an example. The offset may range from 0 ms to 60 ms and is 40 ms in one example, with no limitation intended. Accordingly, the upper limit of true tachyarrhythmia evidence interval criteria applied at block **312** may be 360 ms to 440 ms as examples. The lower limit may be set based on a programmed detection interval or set to a predetermined value. The lower limit may be between 150 ms and 220 ms and is 190 ms in one example. Event intervals shorter than the lower limit may be noise intervals.

(103) When at least one suspected cardiac event interval is determined at block **308** and none fall within the true tachyarrhythmia evidence interval range at block **312**, control circuit **80** may advance to block **314** to determine whether the buffered maximum amplitudes meet suspected

noise criteria. The determination of whether suspected noise criteria are met may correspond to the techniques described above in conjunction with FIG. 5. For example, the lowest maximum amplitude may be required to be less than a noise threshold amplitude, e.g., less than 2.0 mV, and the ratio of the lowest maximum amplitude to the greatest maximum amplitude may be required to be less than a ratio threshold, e.g., less than 1/6. The lowest maximum amplitude may be a mean or median of two or more of the smallest maximum amplitudes or the single lowest maximum amplitude in the amplitude buffer. When the suspected noise criteria are met at block **314**, a suspected noise flag is set at block **315**, e.g., by moving a one into a suspected noise buffer to enable counting of the number of suspected noise determinations out of the most recent n sensed events. Suspected noise is detected by control circuit **80** in the absence of true tachyarrhythmia interval evidence determined from the buffered maximum amplitudes. The true tachyarrhythmia interval evidence is based at least on the greatest maximum amplitude used to determine the amplitude threshold and used to determine whether at least one suspected event interval meets the true tachyarrhythmia interval evidence. When the suspected noise criteria are unmet at block **314**, the suspected noise flag is not set at block **320**, e.g., by moving a zero into the suspected noise buffer. In the example of flow chart **300**, therefore, suspected noise criteria may be applied to the buffered maximum amplitudes when interval-based tachyarrhythmia evidence criteria are unmet.

(104) When control circuit **80** determines that at least one suspected true cardiac event interval meets the tachyarrhythmia interval evidence criteria at block **312**, control circuit **80** may advance to block **316** to determine if morphology based tachyarrhythmia evidence criteria are met before applying suspected noise criteria at block **314**. At block **316**, control circuit **80** determines at least one morphology metric from at least one cardiac signal segment buffered in response to a cardiac sensed event signal. In some examples, control circuit **80** identifies the greatest maximum amplitude stored in the amplitude buffer. Control circuit **80** may determine the morphology metric at block **316** from the buffered cardiac electrical signal segment from which the greatest maximum amplitude was determined.

(105) Example techniques for determining the morphology metrics at block **316** are described below in conjunction with FIGS. **8-10**. The morphology metrics may include a gross morphology amplitude metric determined from the cardiac electrical signal segment associated with the greatest maximum amplitude, e.g., using the techniques described in conjunction with FIG. **9**. The gross morphology amplitude metric is compared to true tachyarrhythmia morphology evidence criteria at block **318**. When the gross morphology amplitude metric is relatively high, the associated cardiac electrical signal is a relatively large signal which may be a true fibrillation waveform or R-wave as opposed to a narrow noise signal spike or relatively smaller baseline noise signal. Therefore a gross morphology amplitude metric that is greater than a threshold may be determined to be evidence of a true cardiac event as opposed to oversensed noise and is determined to be true tachyarrhythmia evidence by control circuit **80**.

(106) The morphology metrics determined at block **316** may include a gross morphology signal width metric. The gross morphology signal width metric may be determined using the techniques described in conjunction with FIG. **10**. When the signal width metric may also be determined by control circuit **80** from the cardiac signal segment associated with the greatest maximum amplitude stored in the amplitude buffer. When the gross morphology signal width metric is relatively high, the associated cardiac electrical signal is a relatively wide signal that has a higher likelihood of being a true fibrillation wave than a noise signal. As such, control circuit **80** may determine evidence of a true tachyarrhythmia at block **318** in response to a gross morphology signal width metric being greater than a threshold value.

(107) Example criteria applied at block **318**, such as a threshold value applied to a gross morphology amplitude metric and a threshold value applied to a gross morphology signal width metric are described below in conjunction with FIGS. **9** and **10**. In some examples, the true tachyarrhythmia morphology evidence criteria are applied only to morphology metrics determined

from the buffered cardiac signal segment associated with the greatest maximum amplitude in the amplitude buffer. It is to be understood, however, that the true tachyarrhythmia morphology evidence criteria may be applied to morphology metrics determined from more than one buffered cardiac signal segment. For example, the true tachyarrhythmia morphology evidence criteria may be applied to all cardiac electrical signal segments associated with the buffered maximum amplitudes that are greater than the amplitude threshold. When control circuit **80** determines that true tachyarrhythmia morphology evidence criteria are met at block **318**, a suspected noise flag is not set at block **320**. The determination of no suspected noise at block **320** may be made by control circuit **80** without determining whether suspected noise criteria are met by the buffered maximum amplitudes. The maximum amplitudes may satisfy suspected noise criteria; however, a suspected noise determination is not made when true tachyarrhythmia morphology evidence criteria are met. Control circuit **80** does count a suspected noise determination for withholding a tachyarrhythmia detection or therapy when the interval-based and morphology-based tachyarrhythmia evidence criteria are both satisfied, indicating that the true underlying rhythm is likely to be a tachyarrhythmia.

(108) In some examples, when both of the gross morphology amplitude metric and the gross morphology signal width metric are greater than a respective threshold value, true tachyarrhythmia morphology evidence criteria are met at block **318**. Control circuit **80** may move a zero into the suspected noise buffer at block **310**, with or without determining if the buffered maximum amplitudes meet the suspected noise criteria. When at least one gross morphology metric is not greater than a threshold value, true tachyarrhythmia morphology evidence criteria is not met at block **318**. Control circuit **80** may advance to block **314** to determine if the buffered maximum amplitudes meet the suspected noise criteria. If so, a one may be moved into the suspected noise buffer at block **316** (or a suspected noise count may be increased). If not, a zero may be moved into the suspected noise buffer at block **320** (or a suspected noise count may be left at the same value or decreased).

(109) FIG. **8** is a diagram **400** of a cardiac electrical signal segment **402** stored over time interval **404** in response to an R-wave sensed event signal **406** and a rectified cardiac electrical signal segment **422**. In some examples, cardiac electrical signal segment **402** is the wideband, notch filtered signal **78** from second sensing channel **85** of FIG. **4**, buffered into memory **82** over time interval **404** in response to an R-wave sensed event signal **406** produced by the first sensing channel **83**. The cardiac electrical signal **402** (e.g., from second sensing channel **85**) may be buffered in memory **82** at a desired sampling rate, e.g., 128 Hz or 256 Hz, until an R-wave sensed event signal **406** is received at which point the desired number of sample points preceding the R-wave sensed event signal **406**, e.g., 68 sample points when the sampling rate is 256 Hz, and the desired number of sample points following the R-wave sensed event signal **406**, e.g., 24 sample points occurring after the R-wave sensed event signal **406**, are stored in memory **82**, in a designated buffer, as the second cardiac electrical signal segment **402** associated with R-wave sensed event signal **406**. In other examples, a higher or lower sampling rate may be used, e.g., a sampling rate of 512 Hz or 128 Hz. A correspondingly higher or lower number of sample points may be buffered to analyze the cardiac signal segment **402** over the same or similar time interval **404**, extending before and after the time point of the R-wave sensed event signal **406**. Time interval **404** over which cardiac electrical signal segment **402** is stored for analysis may be between 100 ms and 500 ms in length in various examples and is 360 ms in one example.

(110) Control circuit **80** determines the maximum peak to peak amplitude **410** as described above for storing in the amplitude buffer, e.g., with a timestamp of the R-wave sensed event signal **406**. Maximum peak to peak amplitude **410** may be determined by control circuit **80** as the difference between the maximum amplitude sample point **412** and the minimum amplitude sample point **414**. The maximum amplitude sample point **412** and the minimum amplitude sample point **414** may be identified from the sample points spanning a portion of the cardiac electrical signal segment **402**,

e.g., over a time interval **416** that is shorter than the time interval **404**. In one example, time interval **404** is 360 ms and time interval **416** is 188 ms. The maximum peak to peak amplitude **410** is determined from the cardiac electrical signal segment over time interval **416** and gross morphology metrics are determined from the rectified cardiac electrical signal segment **422** spanning the longer time interval **404**.

(111) Control circuit **80** may identify maximum peak to peak amplitude **410** as the greatest maximum amplitude stored in the maximum amplitude buffer. The cardiac electrical signal segment associated with the greatest maximum amplitude may be used by control circuit **80** for determining morphology metrics at block **316** of FIG. 7 or block **114** of FIG. 5. Control circuit **80** may determine the gross morphology amplitude metric and the gross morphology signal width metric from the rectified cardiac electrical signal segment **422** in response to the interval based true tachyarrhythmia evidence criteria being met (e.g., as described in conjunction with FIG. 7) or in response to suspected noise criteria being met (e.g., as described in conjunction with FIG. 5).

(112) FIG. 9 is a flow chart **500** of a method for determining a gross morphology amplitude metric of a cardiac electrical signal segment for detecting a tachyarrhythmia morphology according to one example. With reference to FIGS. 8 and 9, at block **502**, the cardiac electrical signal segment **402** stored on a triggered basis in response to an R-wave sensed event signal **406** may be rectified to produce the rectified signal **422**. In some examples, a 360 ms segment of the notch-filtered second cardiac electrical signal may be rectified by rectifier **75** included in second sensing channel **85** (FIG. 4). At block **502**, the buffered, rectified signal segment may be retrieved by control circuit **80** from memory **82**. In other examples, a wideband and notch-filtered signal segment **402** may be buffered in memory **82**, and control circuit **80** may perform the rectification of the stored signal segment at block **502**.

(113) At block **504**, control circuit **80** determines the maximum absolute amplitude **424** (shown in FIG. 8) of the rectified, notch-filtered signal segment **422**. The maximum absolute amplitude **424** may be determined from among all sample points spanning time interval **404** over which the rectified signal segment **422** is buffered. As described above, a 360 ms segment of the second cardiac electrical signal may include 92 sample points when the sampling rate is 256 Hz, with **24** of the sample points occurring after the R-wave sensed event signal **406** that triggered the storage of the signal segment **402** and 68 sample points extending from the R-wave sensed event signal **406** earlier in time from the R-wave sensed event signal **406**.

(114) At block **506**, control circuit **80** sums the amplitudes of all sample points of the rectified signal segment **422**, which represents the area **430** of the rectified signal segment **422**. At block **508**, control circuit **80** may determine the gross morphology amplitude metric of the signal segment **422** as a normalized rectified amplitude (NRA) based on the maximum absolute amplitude **424** determined at block **504** and the summed sample point amplitudes (area **430**) determined at block **506**. In one example, the NRA is determined as a predetermined multiple or weighting of the summation of all sample point amplitudes of the wideband, notch-filtered and rectified signal segment **422** normalized by the maximum absolute amplitude **424**. For instance, the NRA may be determined as four times the summed amplitudes (area **430**) divided by the maximum absolute amplitude **424**, which may be truncated to an integer value. This NRA may be determined as the gross morphology amplitude metric at block **316** of FIG. 7 (or block **114** of FIG. 5) for detecting evidence of a true tachyarrhythmia R-wave or fibrillation wave based on the sample points spanning the signal segment **422** that extends before and after the R-wave sensed event signal **406**. As represented in the diagram **400** of FIG. 8, the gross morphology amplitude metric determined from the maximum absolute amplitude **424** and area **430** represents an amplitude metric of the rectified cardiac signal waveform **422** that may correspond to a tachyarrhythmia waveform morphology.

(115) For example, the gross morphology amplitude metric determined as the weighted area **430** divided by the maximum amplitude **424** may be inversely correlated to the probability of the signal

segment sample points being at a baseline amplitude during the time interval **404**. The higher the gross morphology amplitude metric is, the lower the probability that the signal is at a baseline amplitude at any given time point during the time interval **404**. A relatively low probability that the signal **422** is at baseline during the time interval **404** may be correlated to a tachyarrhythmia morphology, e.g., a ventricular fibrillation morphology, which may resemble a sinusoidal signal. When the gross morphology amplitude metric exceeds a threshold value the more likely the cardiac electrical signal segment has a tachyarrhythmia morphology. When the gross morphology amplitude metric is less than the threshold value, the higher the probability that the signal is at a baseline amplitude at a given time point during the time interval **404** of the signal segment **422** or is a relatively small signal. A relatively higher probability of a signal sample point being at or near baseline during the time segment **404** may be correlated to a relatively narrow R-wave signal occurring during the signal segment, or no true R-wave being present, with baseline amplitude portions of the signal segment occurring before and after the R-wave sensed event signal. As such, when a tachyarrhythmia morphology is not detected and the suspected noise criteria are met, as described above, a suspected noise detection may be made.

(116) When the gross morphology amplitude metric is greater than a predetermined threshold, “yes” branch of block **510**, evidence of a tachyarrhythmia morphology is detected at block **516**. In this case, detection of the tachyarrhythmia morphology may preclude determining suspected noise as described above. When the NRA is less than or equal to the NRA threshold (“no” branch of block **510**), a tachyarrhythmia morphology is not detected at block **512** by control circuit **80**. Control circuit **80** may determine that true tachyarrhythmia evidence criteria are unmet and determine suspected noise based on the buffered maximum peak to peak amplitudes. The NRA threshold for detecting true tachyarrhythmia evidence applied at block **510** may be set between 100 and 150, and is set to 125 in some examples, such as when 92 sample points are summed and multiplied by a weighting factor of four and normalized by the maximum absolute amplitude. The NRA threshold applied to the gross morphology amplitude metric to detect a tachyarrhythmia morphology in the cardiac electrical signal segment may depend on various factors such as the amplification and number of sample points summed, the multiplication or weighting factor of the summed sample points, etc.

(117) FIG. **10** is a flow chart **550** of a method for detecting a tachyarrhythmia morphology in a cardiac electrical signal segment based on a gross morphology signal width metric according to one example. The process of flow chart **550** may be performed by control circuit **80** for determining a gross morphology signal width metric at block **316** of FIG. **7** or block **114** of FIG. **5**. Blocks **502** and **504** correspond to identically-numbered blocks described above in conjunction with FIG. **9**. The notch-filtered, rectified cardiac electrical signal segment, e.g., signal segment **422** shown in FIG. **8**, determined at block **502** may be used to determine the maximum absolute amplitude **424** of the signal segment at block **504**.

(118) With continued reference to FIGS. **8** and **10**, control circuit **80** determines a pulse amplitude threshold **426** at block **552** based on the maximum absolute amplitude **424** determined at block **504**. This pulse amplitude threshold **426** may be used for identifying a signal pulse having a maximum signal width out of all signal pulses occurring during the time segment **404** of the rectified cardiac electrical signal segment **422**. For example, the pulse amplitude threshold **426** used for determining the gross morphology signal width metric may be set to half the maximum absolute amplitude **424** of the rectified, notch-filtered signal segment **422**.

(119) At block **554**, control circuit **80** identifies signal pulses from the cardiac electrical signal segment and determines the signal width for each signal pulse. Each signal pulse in the signal segment **422** may be identified by first identifying zero or baseline amplitude sample points **440**, **442**, **444**, **446**, **448** and **450** of the rectified signal segment **422**. In other examples, the time of zero crossings of the non-rectified signal segment **402** may be identified to demarcate the signal pulses of the rectified signal segment **422** that occur between zero-crossings. In some instances, a sample

point nearest to zero at a zero crossing may be set to a zero amplitude to define a baseline sample point to demarcate signal pulses. Each signal pulse during time interval **404** is identified as the sample points occurring between two consecutive baseline amplitude sample points, e.g., between points **440** and **442**, between points **442** and **444**, between points **444** and **446**, etc. The signal width of each identified signal pulse is determined as the number of sample points (or corresponding time interval) between a pair of consecutive baseline (e.g., zero) amplitude sample points. For example, a signal pulse **432** identified between baseline points **444** and **446** has a signal width **452**, the time interval (or number of sample points) between baseline points **444** and **446**.

(120) The absolute maximum amplitude of each rectified signal pulse is determined at block **556**. All signal pulses of the rectified signal segment **422** that have an absolute maximum amplitude that is greater than or equal to the pulse amplitude threshold **426** are identified at block **558**. For example, all signal pulses having a maximum amplitude that is at least half the maximum absolute amplitude **424** may be identified at block **558**.

(121) In the example of FIG. **8**, signal pulse **432** may be the only signal pulse identified at block **558** since it is the only signal pulse that has an amplitude greater than the amplitude threshold **426**. In other instances, such as in the presence of non-cardiac noise, multiple signal pulses may occur within the cardiac signal segment **422** having an amplitude greater than the amplitude threshold **426**.

(122) Control circuit **80** determines the maximum signal pulse width at block **560** out of all identified signal pulses having an amplitude greater than the pulse amplitude threshold **426**. This maximum pulse width, e.g., pulse width **452** in FIG. **8**, is determined as the gross morphology signal width metric for detecting a tachyarrhythmia morphology. This maximum signal pulse width **452** is expected to be the pulse width of any underlying true cardiac signal waveform in the cardiac signal segment **422**.

(123) The gross morphology signal width metric determined as the maximum pulse width **452** may be correlated to the probability of the buffered cardiac electrical signal segment **402** having a tachyarrhythmia morphology. For example, a relatively high gross morphology signal width metric may be evidence of a relatively wide ventricular fibrillation wave. Conversely, a relatively low gross morphology signal width metric may be evidence of a relatively narrow, true R-wave occurring during the time interval **404** of the cardiac electrical signal segment **402** or absence of a true cardiac signal. When a relatively wide signal pulse is not detected from the rectified cardiac electrical signal segment **422**, an oversensed non-cardiac noise pulse may be present and may have triggered the buffering of the cardiac signal segment **402**. Without evidence of a true tachyarrhythmia morphology (e.g., a relatively wide signal width), a tachyarrhythmia detection may be withheld to appropriately withhold therapy delivering. When evidence of a true tachyarrhythmia morphology is present, detection of the tachyarrhythmia and therapy should not be withheld.

(124) Control circuit **80** compares the maximum pulse width **452** identified at block **560** to a pulse width threshold at block **562**. In one example, the pulse width threshold is set to 20 sample points when the sampling rate is 256 Hz. When the maximum signal pulse width is less than or equal to the width threshold, control circuit **80** does not detect a tachyarrhythmia morphology at block **566**. A maximum signal pulse width that is less than or equal to the width threshold may correspond to a true, relatively narrow R-wave, e.g., during a sinus rhythm or to a non-cardiac noise pulse. Control circuit **80** may apply suspected noise criteria at block **314** of FIG. **7** when the maximum signal pulse width is less than or equal to the width threshold.

(125) When the maximum pulse width is greater than the width threshold at block **562**, the relatively wide maximum signal pulse width may be detected as evidence of a true tachyarrhythmia waveform morphology at block **564**. Evidence of the tachyarrhythmia morphology in signal segment **402**, which may correspond to the greatest maximum amplitude in the maximum amplitude buffer, may preclude a determination of suspected noise, even when the maximum

amplitudes in the amplitude buffer meet the suspected noise criteria.

(126) In various examples, both the gross morphology amplitude metric and the gross morphology signal width metric may be determined (e.g., at block **316** of FIG. **7** or block **114** of FIG. **5** according to the techniques of FIG. **9** and FIG. **10**) and compared to respective threshold values. In some examples, one of the gross morphology amplitude metric or the gross morphology signal width metric may be required to be less than or equal to the respective threshold value for evidence of a true tachyarrhythmia morphology to not be detected, enabling a suspected noise determination to be made by control circuit **80**. When both of the gross morphology amplitude and the gross morphology signal width are greater than the respective tachyarrhythmia morphology threshold value, evidence of a tachyarrhythmia morphology may be detected in the cardiac electrical signal segment, precluding determination of suspected noise. In other examples, both of the gross morphology amplitude and signal width metric may be required to be less than or equal to the respective threshold value in order to determine evidence of a true tachyarrhythmia morphology is not met and allow determination of suspected noise. If one of the gross morphology metrics, e.g., the amplitude or width metric, is greater than or equal to the respective tachyarrhythmia morphology threshold value, the tachyarrhythmia morphology evidence criteria may be met at block **318** of FIG. **7**, preventing control circuit **80** from determining suspected noise criteria are met.

(127) The gross morphology amplitude metric determined by the method of FIG. **9** and the gross morphology signal width metric determined by the method of FIG. **10** may be used in combination to detect evidence of a true tachyarrhythmia morphology at block **318** of FIG. **7** to prevent detection of noise when the underlying rhythm is likely to be a true tachyarrhythmia. A segment of the cardiac electrical signal that has a relatively high gross morphology amplitude metric and/or relatively high gross morphology signal width metric is evidence of a true tachyarrhythmia morphology and prevents noise detection that may lead to withholding of a tachyarrhythmia detection and subsequent therapy in order to maintain a high sensitivity to tachyarrhythmia detection.

(128) FIG. **11** is a flow chart **600** of a method for controlling tachyarrhythmia detection and therapy by a medical device according to one example. At block **602**, control circuit **80** of ICD **14** may determine if criteria for enabling signal analysis for determining noise corruption are met. In one example, the criteria for enabling analysis for determining noise corruption requires counting a threshold number of tachyarrhythmia intervals by control circuit **80**. For instance, a threshold number of VT and/or VF intervals may trigger signal analysis for determination of noise corruption, where the threshold number is less than the NID required to detect VT or VF. For instance, a count of three, five, eight or another selected number of RRIs determined by timing circuit **90** that fall into the VT and/or VF interval zones may be required to enable signal analysis for noise corruption determination at block **602**.

(129) When the criteria for enabling analysis for determining noise corruption are met at block **602**, control circuit **80** buffers a cardiac electrical signal segment at block **608** in response to the currently sensed R-wave (block **606**). Control circuit **80** determines the maximum peak to peak amplitude from the buffered signal segment at block **608**. In some examples, the cardiac electrical signal segment is buffered from the second cardiac electrical signal from second sensing channel **85** when the R-wave is sensed by the first sensing channel **83** from the first cardiac electrical signal. In other examples, the cardiac electrical signal segment may be buffered from the same cardiac electrical signal from which the R-wave was sensed. The maximum peak to peak amplitude is stored with an associated timestamp and/or RRI.

(130) At block **610**, control circuit **80** may determine gross morphology metrics from the buffered signal segment to store in conjunction with the associated maximum peak to peak amplitude. In this way the data for determining noise corruption is available from each buffered signal segment if needed. The buffered signal segment may be overwritten by the next buffered signal segment with

just the maximum peak to peak amplitude, gross morphology metrics and timestamp or RRI stored in respective data buffers for a desired number of R-wave sensed event signals, e.g., 11 to 13 consecutive R-wave sensed event signals. The gross morphology metrics may be determined by control circuit **80** at block **610** according to the techniques described above in conjunction with FIGS. **8-10**.

(131) At block **612**, control circuit **80** determines if the suspected noise criteria are met or not based on the buffered data. For example, control circuit **80** may first determine suspected true cardiac event intervals between maximum peak to peak amplitudes stored in the amplitude buffer that are greater than a threshold amplitude as described in conjunction with FIG. **7**. Control circuit **80** may determine whether true tachyarrhythmia interval evidence criteria are met. In response to true tachyarrhythmia interval evidence criteria being met, control circuit **80** may determine whether the true tachyarrhythmia morphology evidence criteria are met by the gross morphology metrics buffered in the data buffers in association with at least the greatest maximum peak to peak amplitude stored in the amplitude buffer. In response to both the true tachyarrhythmia interval evidence criteria being met and the true tachyarrhythmia morphology evidence criteria being met, control circuit **80** may determine that suspected noise criteria are not met at block **612**. Control circuit **80** updates the suspected noise count at block **614** accordingly.

(132) When at least one of the true tachyarrhythmia interval evidence criteria or the true tachyarrhythmia morphology criteria are not met, control circuit **80** may determine whether the buffered maximum peak to peak amplitudes meet the suspected noise criteria at block **612**. Based on whether the suspected noise criteria are met by the buffered maximum peak to peak amplitudes, control circuit **80** may update the suspected noise count at block **614**. As described above, control circuit **80** may determine that suspected noise criteria are met when at least one maximum peak to peak amplitude is less than a noise threshold amplitude and the ratio of the lowest maximum peak to peak amplitude to the greatest maximum peak to peak amplitude is less than a ratio threshold. In other examples, a ratio of the greatest post-notch filtered peak to peak amplitude to the associated pre-notch filtered peak to peak amplitude may be compared to a relative noise ratio threshold to determine suspected noise criteria are met at block **612**. In some examples, at least one of the interval-based or morphology-based true tachyarrhythmia evidence criteria is not met by the buffered data in order for the suspected noise criteria to be met. In other examples, both of the interval-based and morphology based true tachyarrhythmia evidence criteria may be required to be unmet in order for the suspected noise criteria to be met at block **612**.

(133) While the flow charts presented herein may suggest a particular order of applying the true tachyarrhythmia evidence criteria and the suspected noise criteria, the processes performed by control circuit **80** for determining when criteria are met or not may be performed in a different order than shown in the example flow charts presented herein. Furthermore, the processes performed for determining when criteria are met may be performed simultaneously in parallel processes to arrive at an outcome for updating the suspected noise count at block **614** based on whether true tachyarrhythmia criteria are met (or not) and suspected noise criteria are met (or not).

(134) An X of Y counter may be implemented in control circuit **80** that is updated with each new determination of suspected noise criteria being met or unmet. For example, a first-in-first-out buffer in memory **82** may set a flag indicating suspected noise criteria are met. The buffer may store a value (e.g., 1=suspected noise criteria met and 0=suspected noise criteria not met) for each of a predetermined number of buffer locations, e.g., six to fifteen locations.

(135) At block **616**, control circuit **80** may determine when one or more tachyarrhythmia detection criteria are met. In some examples, tachyarrhythmia detection criteria applied at block **616** may include an interval-based criterion, such as a required NID being reached by the VT interval counter, VF interval counter, or a combined VT/VF interval counter of tachyarrhythmia detection circuit **92**. In other examples, the tachyarrhythmia detection criteria determined to be met or unmet at block **616** may be based on QRS waveform morphology meeting morphology-based criteria or a

combination of interval or rate-based criteria and morphology-based tachyarrhythmia detection criteria.

(136) When no tachyarrhythmia detection criteria are met at block **616**, control circuit **80** may return to block **602** to continue performing signal analysis for determining suspected noise corruption as long as the criteria for enabling noise corruption determination remains satisfied at block **602**. When the criteria for enabling signal analysis for noise detection becomes unmet (e.g., the VT and/or VF interval count falls below a respective threshold for enabling noise corruption determination), the counter or buffer of the suspected noise determinations may be cleared at block **604**. For example, a buffer storing zeros and ones corresponding to suspected noise criteria being unmet or met may be cleared to all zeros at block **604**. After clearing the suspected noise count, control circuit **80** returns to block **602** to wait for the noise corruption determination criteria to be met again.

(137) When at least one criterion for detecting tachyarrhythmia is determined to be satisfied at block **616**, control circuit **80** may compare the current suspected noise count to withhold criteria at block **618**. For example, when the number of suspected noise determinations reaches or exceeds a withhold threshold, the withhold criteria may be met, and the tachyarrhythmia detection based on at least one satisfied tachyarrhythmia detection criterion is withheld by control circuit **80** at block **620**. VT or VF is not detected by control circuit **80** at block **620** when the NID is reached, for example, and a threshold number of suspected noise determinations have been made out of the most recent Y sensed event signals. In some examples, a single suspected noise determination may meet withhold criteria. In other examples, at least three out of twelve buffered values in a suspected noise buffer may be required to be ones, indicating suspected noise criteria have been met three times out of the most recent twelve times the buffered data has been analyzed for determining suspected noise.

(138) When the withhold criteria are met, a tachyarrhythmia therapy that is programmed to be delivered in response to VT or VF detection is not scheduled or delivered. In other examples, the VT or VF detection may be made in response to tachyarrhythmia detection criteria being satisfied at block **616**, but the therapy may be withheld at block **620** when the withhold criteria are met at block **618**. Control circuit **80** may return to block **602** to continue buffering data and analyzing the data for noise corruption as long as the noise corruption determination criteria are met at block **602**.

(139) When the withhold criteria are not met at block **618**, e.g., when the suspected noise count is less than a withhold threshold, the tachyarrhythmia is detected at block **622** by control circuit **80** based on the tachyarrhythmia detection criterion being met at block **616**. Therapy, e.g., ATP and/or CV/DF shock, may be delivered in response to the tachyarrhythmia detection at block **622**. Control circuit **80** may advance to block **604** to reset the suspected noise count after detecting VT/VF and delivering therapy and wait at block **602** until noise corruption determination criteria are met again.

(140) FIG. **12** is a flow chart **700** of a method performed by ICD **14** for detecting noise corruption of a cardiac signal and rejecting a ventricular tachyarrhythmia detection in response to determining noise corruption according to another example. At blocks **702** and **704**, two different cardiac electrical signals may be sensed by sensing circuit **86**. In some examples, two different sensing electrode vectors may be selected by sensing circuit **86** for receiving a first cardiac electrical signal by a first sensing channel **83** and a second cardiac electrical signal by a second sensing channel **85**, respectively. The two sensing electrode vectors may be selected by switching circuitry included in sensing circuit **86** under the control of control circuit **80**. In some examples, the two sensing electrode vectors are programmed by a user and retrieved from memory **82** by control circuit **80** and passed to sensing circuit **86** as vector selection control signals.

(141) In some examples, the first sensing electrode vector selected for sensing the first cardiac electrical signal at block **702** may be a relatively short bipole, e.g., between electrodes **28** and **30** or between electrodes **28** and **24** of lead **16** or other electrode combinations as described above. The relatively short bipole may include electrodes that are in relative close proximity to each other and

to the ventricular heart chambers to provide sensing of a relatively “near-field” ventricular signal for sensing R-waves compared to a second sensing vector selected at block **704**. The first sensing electrode vector may be a vertical sensing vector (with respect to an upright or standing position of the patient) or approximately aligned with the cardiac axis for maximizing the amplitude of R-waves in the first cardiac electrical signal for reliable R-wave sensing. The first sensing electrode vector, however, is not limited to any particular interelectrode spacing or orientation and may be selected as any available electrode pair. The first cardiac electrical signal sensed at block **702** may be a narrow band filtered signal for enhancing the quality of R-wave signals in the first cardiac electrical signal.

(142) The second sensing electrode vector used to receive a second cardiac electrical signal at block **704** may be a relatively longer bipole having an inter-electrode distance that is greater than the first sensing electrode vector. For example, the second sensing electrode vector may be selected as the vector between one of the pace sense electrodes **28** or **30** and ICD housing **15**, one of defibrillation electrodes **24** or **26** and housing **15** or other combinations of one electrode along the distal portion of the lead **16** and the housing **15**. This second sensing electrode vector may be orthogonal or almost orthogonal to the first sensing electrode vector in some examples, but the first and second sensing vectors are not required to be orthogonal vectors. The second sensing electrode vector may receive a relatively more global or far-field cardiac electrical signal compared to the first sensing electrode vector. The second cardiac electrical signal sensed by the second sensing channel **85** at block **704** may be analyzed by control circuit **80** for detecting noise corruption of both of the first and second cardiac electrical signals based on the analysis of only the second cardiac electrical signal. The second cardiac electrical signal may be a wide band filtered signal.

(143) In other examples, the first and second cardiac electrical signals sensed at blocks **702** and **704** may be received from the same sensing electrode vector, such that a single cardiac electrical signal is received by the sensing circuit **86**, but the raw, received signal may be processed by two different sensing channels **83** and **85** of sensing circuit **86** having different filtering and/or other signal processing features to sense two different cardiac electrical signals, one, e.g., a narrow band filtered signal, used by the first sensing channel **83** for sensing R-waves and one, e.g., a wide band filtered signal, sensed by the second sensing channel **85** for determining noise corruption including performing tachyarrhythmia morphology analysis. In still other examples, a single cardiac signal is used for sensing R-waves and buffered for detecting noise corruption of the cardiac signal.

(144) Sensing circuit **86** may produce an R-wave sensed event signal at block **706** in response to the first sensing channel **83** detecting an R-wave sensing threshold crossing by the first cardiac electrical signal. The R-wave sensed event signal may be passed to control circuit **80**. In response to the R-wave sensed event signal, down-going “yes” branch of block **706**, control circuit **80** is triggered at block **708** to store a segment of the second cardiac electrical signal received from the second sensing channel **85** over a predetermined time interval. Segments of the second cardiac electrical signal may be stored in a circulating buffer of memory **82** configured to store multiple sequential segments, where storage of each segment is triggered by an R-wave sensed event signal produced by the first sensing channel **83**. A digitized segment of the second cardiac electrical signal may be 100 to 500 ms long, for example, including sample points before and after the time of the R-wave sensed event signal. The segment of the second cardiac electrical signal may or may not be centered in time on the R-wave sensed event signal received from sensing circuit **86**. For instance, the segment may extend 100 ms after the R-wave sensed event signal and be 200 to 500 ms in duration such that the segment extends from about 100 to 400 ms before the R-wave sensed event signal to 100 ms after. In other examples, the segment may be centered on the R-wave sensed event signal or extend a greater number of sample points after the R-wave sensed event signal than before. In one example, the buffered segment of the cardiac electrical signal is at least 48 sample points obtained at a sampling rate of 256 Hz, or about 188 ms. In another example, the buffered segment is at least 92 sample points, or approximately 360 ms, sampled at 256 Hz.

(145) Memory **82** may be configured to store a predetermined number of second cardiac electrical segments, e.g., at least 1 and in some cases two or more cardiac electrical signal segments, in circulating buffers such that the oldest segment is overwritten by the newest segment. However, previously stored segments may never be analyzed for determining noise corruption and may be overwritten if an R-sense confirmation threshold is not reached at block **714** as described below. In some examples, at least one segment of the second cardiac electrical signal may be stored and if not needed for detecting noise (before noise corruption determination criteria are met), the segment is overwritten by the next segment corresponding to the next R-wave sensed event signal.

(146) In addition to buffering a segment of the second cardiac electrical signal, control circuit **80** responds to the R-wave sensed event signal produced at block **706** by determining an RRI at block **710** ending with the current R-wave sensed event signal and beginning with the most recent preceding R-wave sensed event signal. The timing circuit **90** of control circuit **80** may pass the RRI timing information to the tachyarrhythmia detection circuit **92** which adjusts tachyarrhythmia interval counters at block **712**. If the RRI is longer than a tachycardia detection interval (TDI), the tachyarrhythmia interval counters remain unchanged. If the RRI is shorter than the TDI but longer than a fibrillation detection interval (FDI), e.g., if the RRI is in a tachycardia detection interval zone, a VT interval counter is increased at block **712**. If the RRI is shorter than or equal to the FDI, a VF interval counter is increased at block **712**. In some examples, a combined VT/VF interval counter is increased if the RRI is less than the TDI.

(147) After updating the tachyarrhythmia interval counters at block **712**, tachyarrhythmia detector **92** compares the counter values to an R-sense confirmation threshold at block **714** to determine if noise corruption determination criteria are met. Tachyarrhythmia detector **92** also compares the counter values to VT and VF detection thresholds at block **732** to determine if a respective NID is met. If a VT or VF detection interval counter has reached an R-sense confirmation threshold, “yes” branch of block **714**, control circuit **80** may begin analysis of the second cardiac electrical signal to determine if noise corruption is likely causing false R-wave sensed event signals to be produced by the first sensing channel **83**, resulting in VT and/or VF counters being increased at block **712**. The R-sense confirmation threshold may be a VT or VF interval count value that is greater than one or another higher threshold count value, indicating that a tachycardia may be suspected but has not yet reached a detection threshold. Different R-sense confirmation thresholds may be applied to the VT interval counter and the VF interval counter. For example, the R-sense confirmation threshold may be a count of two on the VT interval counter and a count of three on the VF interval counter. In other examples, the R-sense confirmation threshold is a higher number, for example five or higher, but may be less than the number of VT or VF intervals required to detect VT or VF. In addition or alternatively to applying an R-sense confirmation threshold to the individual VT and VF counters, an R-sense confirmation threshold may be applied to a combined VT/VF interval counter. It is recognized that in some examples, VT detection may not be enabled and VF detection may be enabled. In this case, only a VF interval counter is updated at block **712** in response to RRI determinations, and the R-sense confirmation threshold may be applied to the VF interval counter at block **714**.

(148) If the R-sense confirmation threshold is not reached by any of the tachyarrhythmia interval counters at block **714**, the control circuit **80** waits for the next R-wave sensed event signal at block **708** to buffer the next segment of the second cardiac electrical signal. If the R-sense confirmation threshold is reached at block **714**, e.g., when the VF interval counter is greater than 2, the control circuit **80** begins analysis of the buffered second cardiac electrical signal segments for detecting noise corruption.

(149) At block **716**, control circuit **80** may notch filter the buffered signal segment. In other examples, the second sensing channel **85** may include a notch filter such that a notch filtered signal segment is buffered at block **708**. However, both the wide band filtered signal prior to notch filtering and the notch filtered signal may be used in the analysis for determining noise corruption

in some examples. Both the wide band filtered signal and the wideband and notch filtered signal may be buffered at block **708** or the wide band filtered signal may be buffered and subsequently notch filtered at block **716** using hardware, firmware or software. As described above, the notch-filter may be implemented to attenuate 50 Hz and/or 60 Hz line frequency noise (and in some cases harmonics thereof).

(150) At block **718**, control circuit **80** determines the maximum peak to peak amplitude from at least the notch filtered signal segment. In some examples, the maximum peak to peak amplitude is determined from a portion of the notch filtered signal segment. For example, if the signal segment buffered at block **708** is 300 ms or longer, e.g., 360 ms, the maximum peak to peak amplitude may be determined from a portion of the signal segment encompassing the time of the associated R-wave sensed event signal, e.g., from a 150 to 200 ms segment. In one example, the maximum peak to peak amplitude is determined as the difference between the maximum sample point amplitude and the minimum sample point amplitude that occur during a 188 ms segment that encompasses the time of the associated R-wave sensed event signal.

(151) In some examples, control circuit **80** may determine the maximum peak to peak amplitude from the post-notch filtered signal segment and from the pre-notch filtered signal segment at block **718**. The determination of both enables control circuit **80** to determine a ratio of the post- to pre-notch filtered maximum peak to peak amplitudes for comparison to a relative noise threshold for detecting suspected noise at block **730**. In other examples, only the maximum peak to peak amplitude of the notch filtered segment is determined at block **718**.

(152) At block **720**, control circuit **80** may determine the gross morphology metric(s) for determining when tachyarrhythmia morphology evidence criteria are met at block **728**. The gross morphology metrics may be determined as generally described in conjunction with FIGS. **9** and **10** above. The post-notch filtered maximum peak to peak amplitude, the timestamp or associated RRI, the gross morphology metrics, and in some examples the pre-notch filtered maximum peak to peak amplitude may all be stored in data buffers in memory **82**. Control circuit **80** may determine if the required minimum number of buffered maximum peak to peak amplitudes have been stored at block **722** to proceed with the noise corruption determination. If not, control circuit **80** returns to block **708** to obtain additional cardiac electrical signal segment data as long as the R-sense confirmation threshold is met. In some examples, the amplitude buffer must be filled with at least three maximum peak to peak amplitudes in order to advance to determine noise corruption. If the minimum is met at block **722**, control circuit **80** advances to block **724**.

(153) At block **724**, control circuit **80** determines suspected true event intervals based on the buffered maximum peak to peak amplitudes. Determination of whether true tachyarrhythmia interval criteria are met is based at least on the greatest maximum peak to peak amplitude stored in the amplitude buffer. As described above in conjunction with FIG. **7**, control circuit **80** may identify buffered maximum peak to peak amplitudes that are greater than a threshold amplitude set based on the greatest maximum peak to peak amplitude. Control circuit may determine one or more suspected true event intervals between the identified maximum peak to peak amplitudes. When at least one suspected true event interval falls within a specified interval range, or when only a single buffered maximum peak to peak amplitude is greater than the threshold amplitude, control circuit **80** may determine that true tachyarrhythmia interval evidence criteria are met at block **726**. If at least one suspected true event interval is determined and no suspected true event intervals fall within the specified interval range, control circuit **80** may advance directly to block **730** to determine if suspected noise criteria are met.

(154) When the true tachyarrhythmia interval criteria are met at block **726**, control circuit **80** determines whether the buffered gross morphology metrics meet true tachyarrhythmia morphology criteria at block **728**. As described above, the gross morphology metric(s) associated with a greatest maximum peak to peak amplitude stored in the maximum amplitude buffer may be compared to respective threshold(s). For example, when both of the gross morphology amplitude metric and the

gross morphology signal width metric associated with the greatest maximum peak to peak amplitudes are greater than a respective threshold, true tachyarrhythmia morphology criteria are met at block **728**. Control circuit **80** advances to block **731** to update a suspected noise count (e.g., by moving a zero into a suspected noise buffer) indicating that suspected noise is not determined based on the currently buffered maximum peak to peak amplitudes and associated data. Since the true tachyarrhythmia interval criteria and the true tachyarrhythmia morphology criteria are both satisfied, suspected noise is not determined based on the currently buffered data without comparing the buffered maximum peak to peak amplitudes to the suspected noise criteria at block **730**.

(155) Control circuit **80** may determine that at least one of the gross morphology amplitude metric or the gross morphology signal width metric is less than or equal to a respective threshold at block **728**. In this case, control circuit **80** may determine that the true tachyarrhythmia morphology criteria are not met. When true tachyarrhythmia interval criteria are met (block **726**) but true tachyarrhythmia morphology criteria are unmet, control circuit **80** applies suspected noise criteria to the buffered maximum peak to peak amplitude data at block **730**. In the example shown, the true tachyarrhythmia interval criteria is required to be met at block **726** before applying the true tachyarrhythmia morphology criteria at block **728**. In other examples, however, control circuit **80** may require that the true tachyarrhythmia morphology criteria be met before checking if the true tachyarrhythmia interval criteria are met. In still other examples, both the true tachyarrhythmia interval and morphology criteria may be applied and if either are unmet, suspected noise criteria may be applied at block **730**. In still other examples, control circuit **80** may determine whether the suspected noise criteria are met at block **730** and, if met, control circuit **80** may verify that at least one of the true tachyarrhythmia interval criteria or the true tachyarrhythmia morphology criteria are unmet before detecting suspected noise based on the buffered maximum peak to peak amplitudes. The criteria applied at blocks **726**, **728** and **730**, therefore, may be applied in a different order than shown or applied in parallel in order to determine whether to increase the suspected noise count at block **731** or not.

(156) At block **730**, control circuit **80** compares buffered maximum peak to peak amplitudes to the suspected noise criteria. As described above, the suspected noise criteria may require that a lowest maximum peak to peak amplitude (which may be determined as the mean or median of two or more smallest maximum peak to peak amplitudes) is less than a noise threshold amplitude. Additionally or alternatively, the suspected noise criteria may require that the ratio of the lowest maximum peak to peak amplitude to the greatest maximum peak to peak amplitude is less than a threshold ratio. Additionally or alternatively, the suspected noise criteria may require that a ratio of the post-notch filtered maximum peak to peak amplitude to the pre-notch filtered maximum peak to peak amplitude for at least one buffered maximum amplitude is less than a relative noise threshold.

(157) When control circuit **80** determines that the buffered maximum peak to peak amplitude data meets the suspected noise criteria at block **730**, control circuit **80** increases a suspected noise count at block **731**. As described above, the noise count may be increased by moving a one into a suspected noise buffer configured to store a predetermined number of ones or zeros for facilitating counting the number of times suspected noise criteria are met over the most recent, predetermined number of R-wave sensed event signals. In some examples, the suspected noise counter is configured to store up to twelve values, ones or zeros, to facilitate determining the number of times the suspected noise criteria are met over the most recent twelve R-wave sensed event signals. It is noted, that the current R-wave sensed event signal may or may not be a suspected noise signal. The determination of noise criteria being met at block **730** may be based on any of the buffered maximum peak to peak amplitudes meeting the suspected noise criteria.

(158) After updating the suspected noise count at block **731**, control circuit **80** may determine if the NID has been reached by the VT, VF or combined VT/VF interval counters at block **732**. When a threshold number of intervals to detect (NID) is not reached by the VT interval counter, VF interval counter, or combined VT/VF interval counter, control circuit **80** returns to block **710** to continue

determining RRI and analyzing segment cardiac electrical signal segments as long as the R-sense confirmation threshold is satisfied (block **714**). If the R-sense confirmation threshold is no longer met at block **714**, the suspected noise counter may be cleared at block **715**, e.g., by setting all values in a suspected noise buffer to zeros.

(159) When the NID is reached at block **732**, based on the values of the VT and/or VF interval counters, control circuit **80** determines whether a withhold detection threshold number of suspected noise determinations has been reached (or exceeded) at block **734**. The VT or VF detection based on the NID being reached is withheld at block **736** in response to a withhold threshold number of suspected noise determinations. In one example, if at least three out of twelve values in the suspected noise buffer are one, indicating at least three suspected noise determinations based on the buffered maximum peak to peak amplitude data over the most recent twelve R-wave sensed event signals, control circuit **80** determines noise corruption, and the withhold criteria are met at block **734**. The VT or VF detection (and any associated VT or VF therapy) based on the NID being reached is withheld by control circuit **80** at block **736**. While not explicitly shown by the flow chart **700** for the sake of clarity, as long as the NID continues to be met at block **732**, control circuit **80** may continue to determine whether suspected noise criteria are met at blocks **716** through **730** as new R-waves are sensed and update the suspected noise count at block **731**.

(160) In some examples, control circuit **80** may determine if termination criteria are met at block **738** when detection has been withheld based on the determination of noise corruption (withhold criteria met at block **734**). Termination of the fast rhythm may be detected based on a predetermined number of RRIs that are greater than a tachyarrhythmia detection interval or when a mean, median or other metric of RRIs determined over predetermined time interval is greater than a tachyarrhythmia detection interval. For example, when a threshold number of RRIs longer than the VT detection interval (e.g., when VT detection is enabled) or longer than the VF detection interval (e.g., when VT detection is not enabled) are detected subsequent to the NID being met, tachyarrhythmia termination may be detected at block **738**. In one example, termination is detected at block **738** when at least eight consecutive long RRIs, e.g., greater than the VT detection interval, are detected. In another example, control circuit **80** may detect termination at block **738** when a predetermined time interval elapses and a median RRI is greater than the VT detection interval. For instance, when the median RRI of the most recent 12 RRIs is always greater than the VT detection interval for at least 20 seconds, or other predetermined time period, control circuit **80** may detect termination at block **738**. Control circuit **80** may reset the VT and VF interval counters and the suspected noise count and return to block **710** in response to detecting termination.

(161) When the NID is met at block **732** and the withhold criteria are not met at block **734**, e.g., due to the suspected noise count being less than a threshold number, a VT or VF episode is detected at block **740**. Therapy delivery circuit **84** may deliver a VT or VF therapy at block **742** in response to the VT/VF detection. It is to be understood that other criteria besides the NID criterion may be applied by control circuit **80** before detecting the VT or VF at block **740**. For example, various P-wave oversensing rejection criteria, T-wave oversensing rejection criteria, supraventricular tachycardia (SVT) rejection criteria, etc. may be required to be unmet and/or tachyarrhythmia onset criteria, tachyarrhythmia morphology criteria, or other tachyarrhythmia criteria may be required to be met before detecting VT/VF at block **740** and delivering therapy at block **742**.

(162) It is contemplated that in other examples, the VT/VF detection may be made in response to detection criteria being satisfied, e.g., the NID being reached at block **732**, but the VT/VF therapy may be withheld at block **736** when the withhold criteria is met or exceeded by the suspected noise count. Therapy delivery circuit **84** may withhold a VT or VF therapy until the withhold criteria is no longer met and the tachyarrhythmia is still being detected. Therapy delivery circuit **84** may deliver a withheld therapy when the withhold criteria are no longer met and termination of the detected VT or VF has not been detected at block **738**. If the detected VT or VF is determined to be terminated at block **738** while the therapy is being withheld, the suspected noise count and VT/VF

interval counters may all be cleared, and the process may return to block 710 without ever delivering a therapy.

(163) It should be understood that, depending on the example, certain acts or events of any of the methods described herein can be performed in a different sequence, may be added, merged, or left out altogether (e.g., not all described acts or events are necessary for the practice of the method). Moreover, in certain examples, acts or events may be performed concurrently, e.g., through multi-threaded processing, interrupt processing, or multiple processors, rather than sequentially. In addition, while certain aspects of this disclosure are described as being performed by a single circuit or unit for purposes of clarity, it should be understood that the techniques of this disclosure may be performed by a combination of units or circuits associated with, for example, a medical device.

(164) In one or more examples, the functions described may be implemented in hardware, software, firmware, or any combination thereof. If implemented in software, the functions may be stored as one or more instructions or code on a computer-readable medium and executed by a hardware-based processing unit. Computer-readable media may include computer-readable storage media, which corresponds to a tangible medium such as data storage media (e.g., RAM, ROM, EEPROM, flash memory, or any other medium that can be used to store desired program code in the form of instructions or data structures and that can be accessed by a computer).

(165) Instructions may be executed by one or more processors, such as one or more digital signal processors (DSPs), general purpose microprocessors, application specific integrated circuits (ASICs), field programmable logic arrays (FPLAs), or other equivalent integrated or discrete logic circuitry. Accordingly, the term “processor,” as used herein may refer to any of the foregoing structure or any other structure suitable for implementation of the techniques described herein. Also, the techniques could be fully implemented in one or more circuits or logic elements.

(166) Thus, a medical device has been presented in the foregoing description with reference to specific examples. It is to be understood that various aspects disclosed herein may be combined in different combinations than the specific combinations presented in the accompanying drawings. It is appreciated that various modifications to the referenced examples may be made without departing from the scope of the disclosure and the following claims.

Claims

1. A medical device comprising: a sensing circuit configured to: sense one or more cardiac electrical signals; and sense event signals from the one or more cardiac electrical signals; a therapy delivery circuit configured to deliver a tachyarrhythmia therapy; a control circuit coupled to the sensing circuit and the therapy delivery circuit, the control circuit configured to: for each of a plurality of event signals sensed by the sensing circuit, determine a peak amplitude from one of the one or more cardiac electrical signals; determine a lowest peak amplitude from the determined peak amplitudes; determine a greatest peak amplitude from the determined peak amplitudes; determine at least one suspected true cardiac event interval from the peak amplitudes; determine that the at least one suspected true cardiac event interval is outside a true tachyarrhythmia evidence interval range; in response to the at least one suspected true cardiac event interval being outside the true tachyarrhythmia evidence interval range, apply suspected noise criteria to the determined peak amplitudes for detecting noise corruption; determine that the lowest peak amplitude and the greatest peak amplitude meet the suspected noise criteria; detect noise corruption of the one or more cardiac electrical signals based on at least the lowest peak amplitude and the greatest peak amplitude determined for the plurality of event signals meeting the suspected noise criteria; and control the therapy delivery circuit to withhold the tachyarrhythmia therapy in response to detecting the noise corruption.

2. The medical device of claim 1 wherein the control circuit is further configured to determine the

lowest peak amplitude by: identifying at least two smallest peak amplitude values of the determined peak amplitudes; and determining the lowest peak amplitude based on at least the two smallest peak amplitude values.

3. The medical device of claim 1 wherein the control circuit is further configured to determine that the lowest peak amplitude and the greatest peak amplitude meet the suspected noise criteria by: determining a ratio of the greatest peak amplitude to the lowest peak amplitude; and determining that the ratio is greater than a ratio threshold.

4. The medical device of claim 1 wherein the control circuit is further configured to determine that the lowest peak amplitude and the greatest peak amplitude meet the suspected noise criteria by determining that the lowest peak amplitude is less than a noise amplitude threshold.

5. The medical device of claim 4 wherein the control circuit is further configured to: determine one or more R-wave peak amplitudes from the one or more cardiac electrical signals; and determine the noise amplitude threshold from the one or more R-wave peak amplitudes.

6. The medical device of claim 1 wherein: the sensing circuit is further configured to: sense the one or more cardiac electrical signals by sensing at least a first cardiac electrical signal and a second cardiac electrical signal; and sense the event signals from a first cardiac electrical signal; and the control circuit is further configured to determine the peak amplitude for each of the plurality of event signals sensed by the sensing circuit by determining a peak to peak amplitude from the second cardiac electrical signal.

7. The medical device of claim 1 wherein the control circuit is further configured to: determine at least one tachyarrhythmia metric based on at least the identified greatest maximum amplitude; determine the at least one tachyarrhythmia metric does not meet true tachyarrhythmia evidence criteria; and determine that the suspected noise criteria are met when the at least one tachyarrhythmia metric does not meet the true tachyarrhythmia evidence criteria.

8. The medical device of claim 1 wherein the control circuit is further configured to: identify at least two smallest peak amplitude values of the determined peak amplitudes; determine the lowest peak amplitude based on at least the two smallest peak amplitude values; determine a ratio of the greatest peak amplitude to the lowest peak amplitude; and determine that the lowest peak amplitude and the greatest peak amplitude meet the suspected noise criteria by at least: determining that the ratio is greater than a ratio threshold; and determining that the lowest peak amplitude is less than a noise threshold.

9. The medical device of claim 1 wherein the control circuit is further configured to: detect a threshold number of tachyarrhythmia intervals based on the event signals sensed by the sensing circuit; in response to detecting noise corruption, withhold a tachyarrhythmia detection based on the threshold number of tachyarrhythmia intervals being detected; and control the therapy delivery circuit to withhold the tachyarrhythmia therapy by withholding the tachyarrhythmia detection.

10. The medical device of claim 9 wherein: the control circuit is further configured to: determine that the suspected noise criteria are not met based on a second plurality of cardiac events sensed by the sensing circuit; determine that noise corruption is not detected based on at least the second plurality of cardiac events sensed by the sensing circuit not meeting the suspected noise criteria; and detect a tachyarrhythmia in response to the threshold number of tachyarrhythmia intervals being detected and noise corruption not being detected; and the therapy delivery circuit further configured to deliver the tachyarrhythmia therapy in response to the tachyarrhythmia detection.

11. The medical device of claim 1 wherein the control circuit is further configured to determine that the at least one suspected true cardiac event interval is outside the true tachyarrhythmia evidence interval range by determining that the at least one suspected true cardiac event interval is less than a lower limit of the true tachyarrhythmia evidence interval range.

12. A method comprising: sensing one or more cardiac electrical signals; sensing event signals from the one or more cardiac electrical signals; for each of a plurality of sensed event signals, determining a peak amplitude from one of the one or more cardiac electrical signals; determining a

lowest peak amplitude from the determined peak amplitudes; determining a greatest peak amplitude from the determined peak amplitudes; determining at least one suspected true cardiac event interval from the peak amplitudes; determining that the at least one suspected true cardiac event interval is outside a true tachyarrhythmia evidence interval range; in response to the at least one suspected true cardiac event interval being outside the true tachyarrhythmia evidence interval range, applying suspected noise criteria to the determined peak amplitudes for detecting noise corruption; determining that the lowest peak amplitude and the greatest peak amplitude meet suspected noise criteria; detecting noise corruption of the one or more cardiac electrical signals based on at least the lowest peak amplitude and the greatest peak amplitude determined for the plurality of event signals meeting the suspected noise criteria; and withholding a tachyarrhythmia therapy in response to detecting the noise corruption.

13. The method of claim 12 wherein determining the lowest peak amplitude comprises: identifying at least two smallest peak amplitude values of the determined peak amplitudes; and determining the lowest peak amplitude based on at least the two smallest peak amplitude values.

14. The method of claim 12 wherein determining that the lowest peak amplitude and the greatest peak amplitude meet the suspected noise criteria comprises: determining a ratio of the greatest peak amplitude to the lowest peak amplitude; and determining that the ratio is greater than a ratio threshold.

15. The method of claim 12 wherein determining that the lowest peak amplitude and the greatest peak amplitude meet the suspected noise criteria comprises determining that the lowest peak amplitude is less than a noise amplitude threshold.

16. The method of claim 15 further comprising: determining one or more R-wave peak amplitudes from the one or more cardiac electrical signals; and determining the noise amplitude threshold from the one or more R-wave peak amplitudes.

17. The method of claim 12 further comprising: sensing the one or more cardiac electrical signals by sensing at least a first cardiac electrical signal and a second cardiac electrical signal; sensing the event signals from the first cardiac electrical signal; and determining the peak amplitude for each of the plurality of event signals by determining a peak to peak amplitude from the second cardiac electrical signal.

18. The method of claim 12 wherein determining that the lowest peak amplitude and the greatest peak amplitude meet the suspected noise criteria comprises: determining at least one tachyarrhythmia metric based on at least the identified greatest maximum amplitude; determining the at least one tachyarrhythmia metric does not meet true tachyarrhythmia evidence criteria; and determining that the suspected noise criteria are met when the at least one tachyarrhythmia metric does not meet the true tachyarrhythmia evidence criteria.

19. The method of claim 12 further comprising: identifying at least two smallest peak amplitude values of the determined peak amplitudes; determining the lowest peak amplitude based on at least the two smallest peak amplitude values; determining a ratio of the greatest peak amplitude to the lowest peak amplitude; and determining that the lowest peak amplitude and the greatest peak amplitude meet the suspected noise criteria by at least: determining that the ratio is greater than a ratio threshold; and determining that the lowest peak amplitude is less than a noise threshold.

20. The method of claim 12 further comprising: detecting a threshold number of tachyarrhythmia intervals based on the event signals sensed by the sensing circuit; in response to detecting 0.noise corruption, withholding a tachyarrhythmia detection based on the threshold number of tachyarrhythmia intervals being detected; and withholding the tachyarrhythmia therapy by withholding the tachyarrhythmia detection.

21. The method of claim 20 further comprising: determining that the suspected noise criteria are not met based on a second plurality of cardiac events; determining that noise corruption is not detected based on at least the second plurality of cardiac events not meeting the suspected noise criteria; detecting a tachyarrhythmia in response to the threshold number of tachyarrhythmia

intervals being detected and noise corruption not being detected; and delivering the tachyarrhythmia therapy in response to the tachyarrhythmia detection.

22. A non-transitory computer readable medium storing a set of instructions which, when executed by a control circuit of a medical device, cause the medical device to: sense one or more cardiac electrical signals; sense event signals from the one or more cardiac electrical signals; for each of a plurality of sensed event signals, determine a peak amplitude from one of the one or more cardiac electrical signals; determine a lowest peak amplitude from the determined peak amplitudes; determine a greatest peak amplitude from the determined peak amplitudes; determine at least one suspected true cardiac event interval from the peak amplitudes; determine that the at least one suspected true cardiac event interval is outside a true tachyarrhythmia evidence interval range; in response to the at least one suspected true cardiac event interval being outside the true tachyarrhythmia evidence interval range, apply suspected noise criteria to the determined peak amplitudes for detecting noise corruption; determine that the lowest peak amplitude and the greatest peak amplitude meet the suspected noise criteria; detect noise corruption of the one or more cardiac electrical signals based on at least the lowest peak amplitude and the greatest peak amplitude determined for the plurality of event signals meeting the suspected noise criteria; and withhold a tachyarrhythmia therapy in response to detecting the noise corruption.
